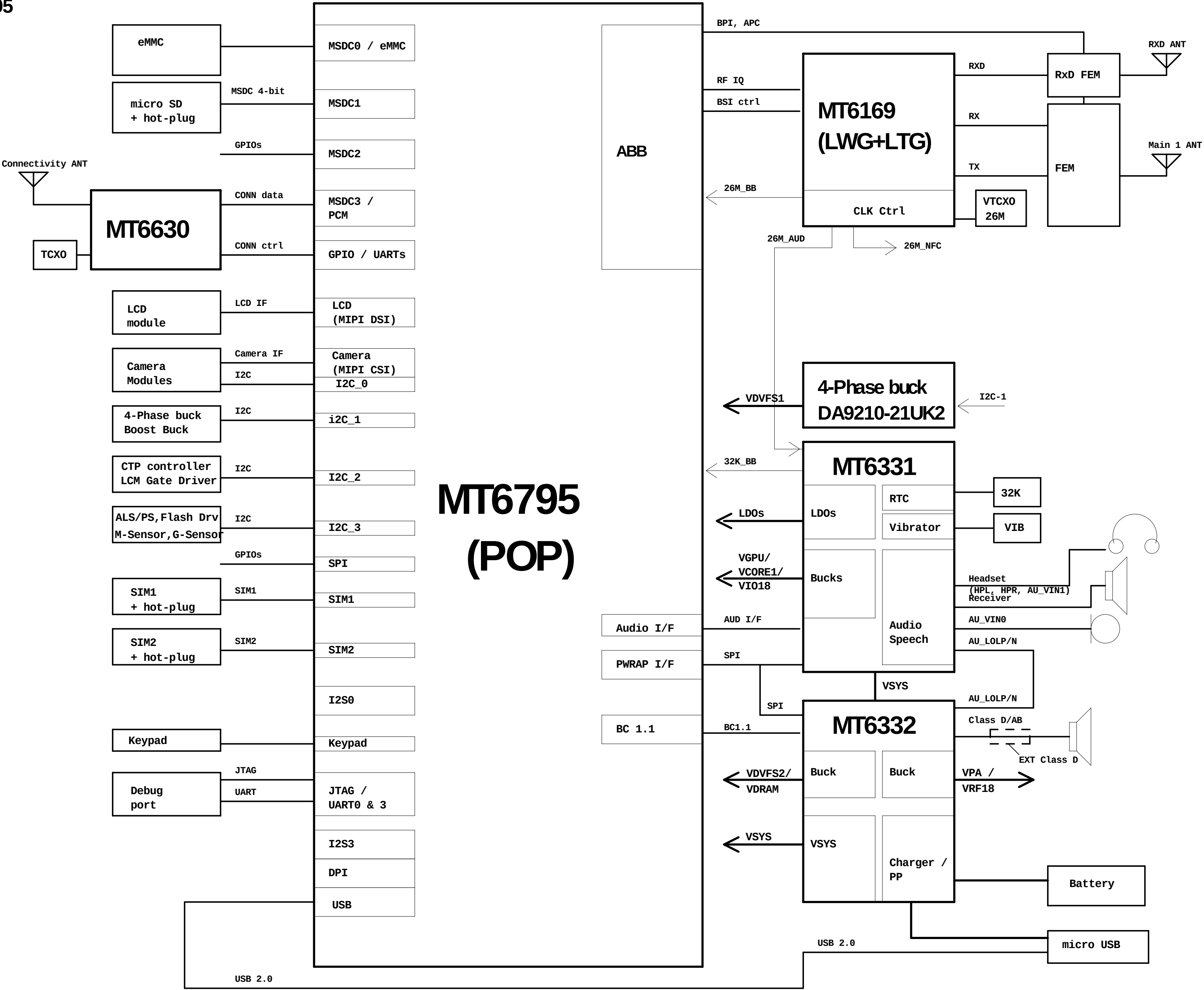




REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 00_BLOCK_DIAGRAM		VERSION: V1.1	SHEET: 1 OF 22
CHECKED	< >	DATED	< >				

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

I2C	Function	I2C Spec.	Budgeted Timing	I2C Slave Address (7-bit mode)
I2C-0	Rear Camera	400 Kbps	Yes.	Rear camera (IMX220) I2C address: 0X10 (Write:0x20, Read:0x21) AF driver (DW9718S) I2C address: 0X0C (Write:0x18, Read:0x19) EEPROM (GT24C16) I2C address: 0X05 (Write:0xA0, Read:0xA1)
	Front Camera	400 Kbps	Yes.	Front camera (OV5858) I2C address: 0X36 (Write:0x6C, Read:0x6D) AF driver (DW9714) I2C address: 0X0C (Write:0x18, Read:0x19)
I2C-1	4-Phase buck	3.4 Mbps	Yes.	DA9210-21UK2 / 4-Phase Buck I2C address: 0X6B (Write:0xD6, Read:0xD7)
	VBAT Boost	3.4 Mbps		TPS61280 / Boost buck I2C address: 0X75 (Write:0xEA, Read:0xEB)
I2C-2	CTP	400 Kbps	Yes.	GT910/ CTP I2C address: 0X14 (Write:0x28, Read:0x29)
	LCM Gate Driver	400 Kbps	Yes.	NT50358CG/J/ CTP I2C address: 0X3E (Write:0x7C, Read:0x7D)
I2C-3	M Sensor	400 Kbps		AK8963C / M-Sensor I2C Address: 0x0C (Write:0x18, Read:0x19) MMC35240P/J/ M-Sensor I2C Address: 0x30 (Write:0x60, Read:0x61)
	A+Gyro Sensor	400 Kbps	Yes.	x
	G-Sensor	400 Kbps		KXTJ2-1009 / M-Sensor I2C Address: 0x0E (Write:0x1C, Read:0x14) BMA222E/ G-Sensor I2C Address: 0x15(Write:0x30, Read:0x31)
	ALS/ PS Sensor	400 Kbps		LTR-559ALS-034/ ALS+PS I2C address: 0X23 (Write:0x46, Read:0x47)
	NFC (x)	1.2 Mbps	Yes.	x
	MHL (x)	400 Kbps		x
	FLASH DRV	400 Kbps		LM3643YFFR/FLASH DRV I2C address: 0X63 (Write:0xC6, Read:0xC7)
	Smart PA	400 Kbps		x

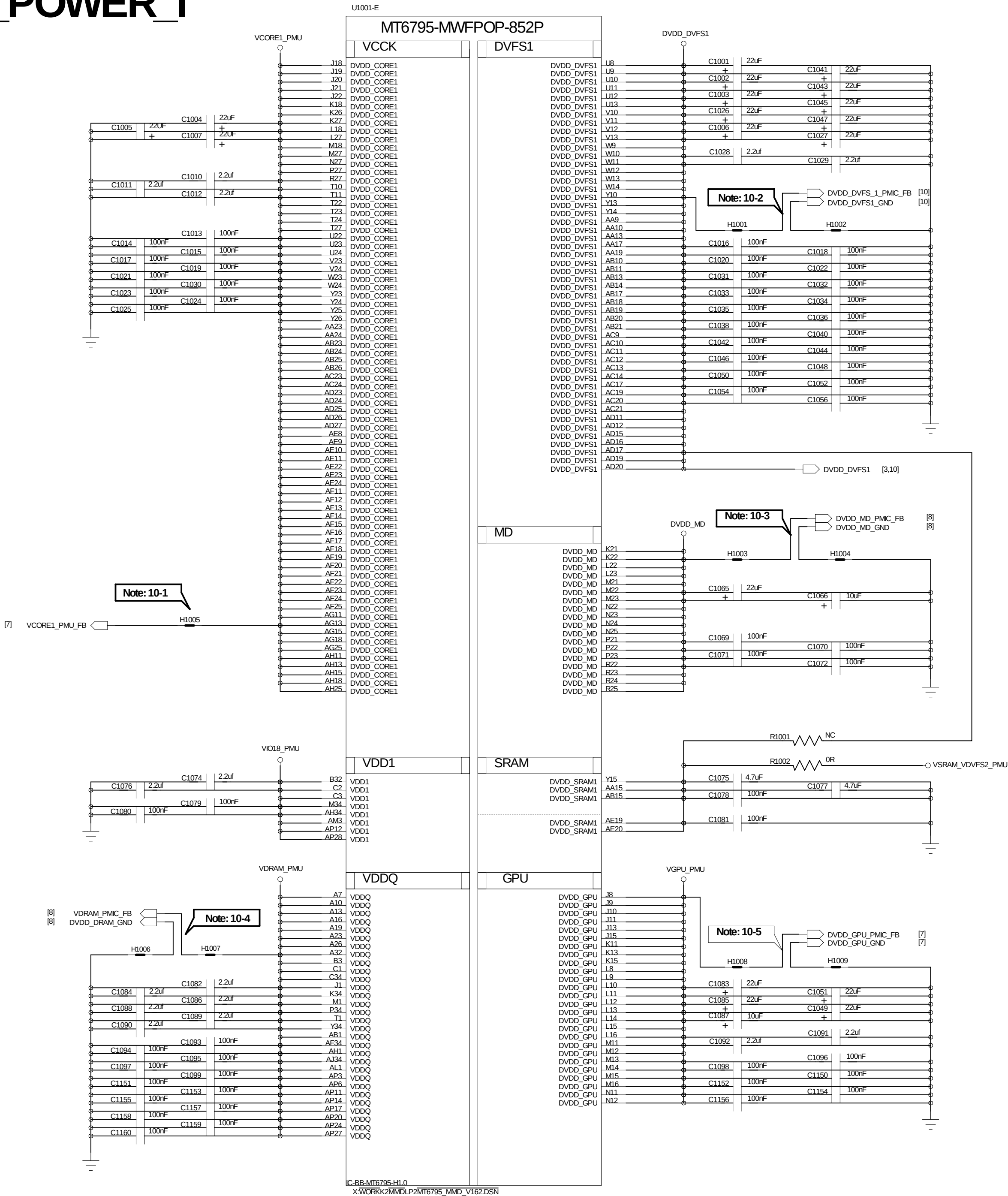
Note 1: I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

Note 2: It is recommended to use I2C-3 as the I2C control interface for front camera application that needs to support auto-focus (AF) or PIP/VIV feature.
I2C control interface of front camera (with AF) needs to be assigned to I2C-3 bus in order to deal with the case of the I2C slave address of front camera's AF is identical with rear camera's AF to avoid I2C bus conflict.

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952	Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 01_I2C_ID_OVERVIEW	VERSION: V1.1	SHEET: 2 OF 22
CHECKED	< >	DATED	< >			

BB_POWER_I

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



Schematic design notice of "10_BB_POWER_1" page.

- Note 10-1: VCORE_1 remote sense must be close to MT6795i | s AG13 ball.
Remote sense trace with GNDshielding to PMIC (Differential)
- Note 10-2: Differential pair of DVFS1 remote sense must be close to MT6795i | s Y10 ball.
- Note 10-3: Differential pair of DVDD_MD remote sense must be close to MT6795i | s K22 ball.
- Note 10-4: VDRAM remote sense must be close to MT6795i | s A32 ball or be placed to the farrest power ball of VDRAM from MT6332 point of view.
- Note 10-5: Differential pair of GPU remote sense must be close to MT6795i | s J8 ball.

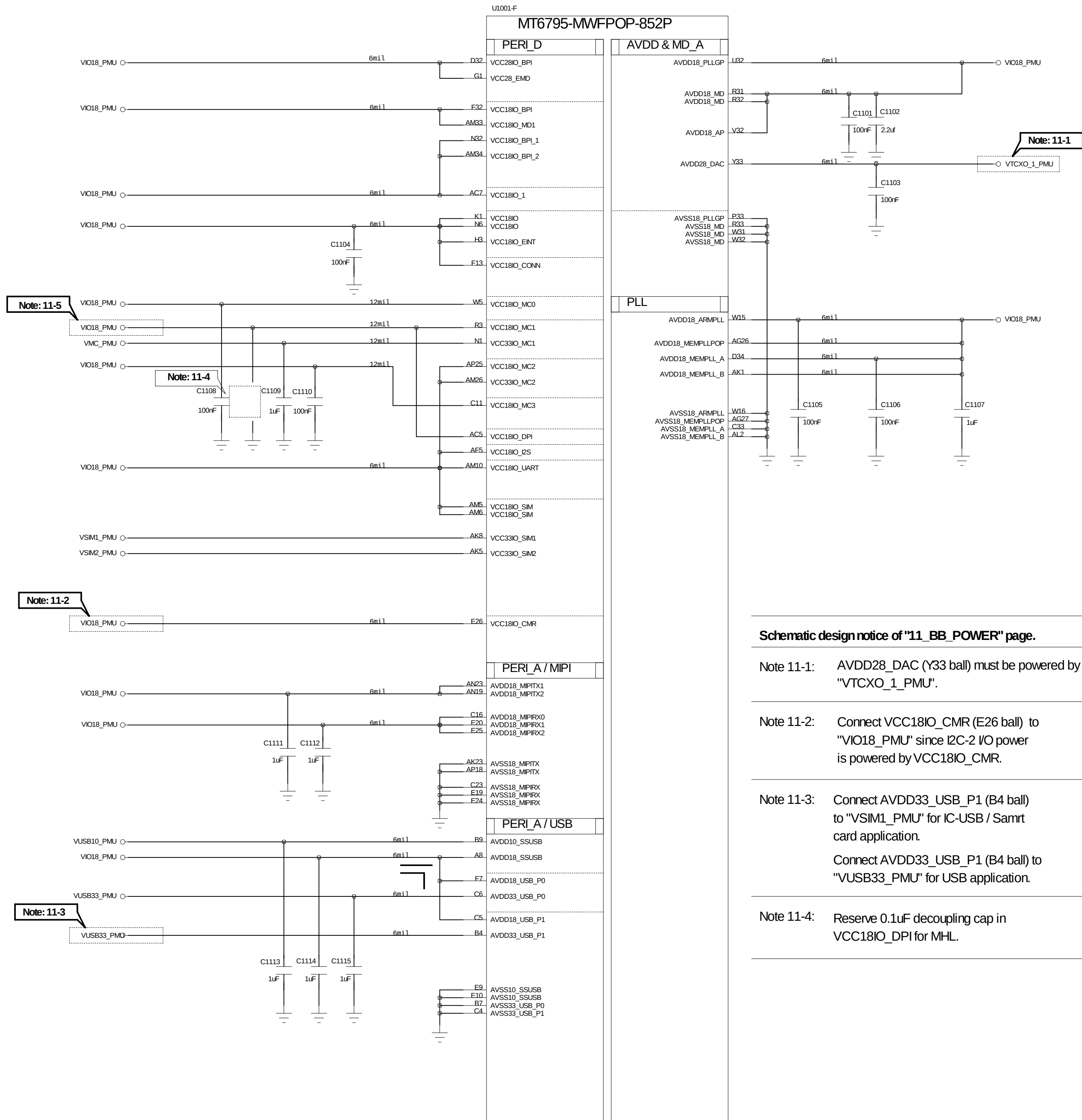
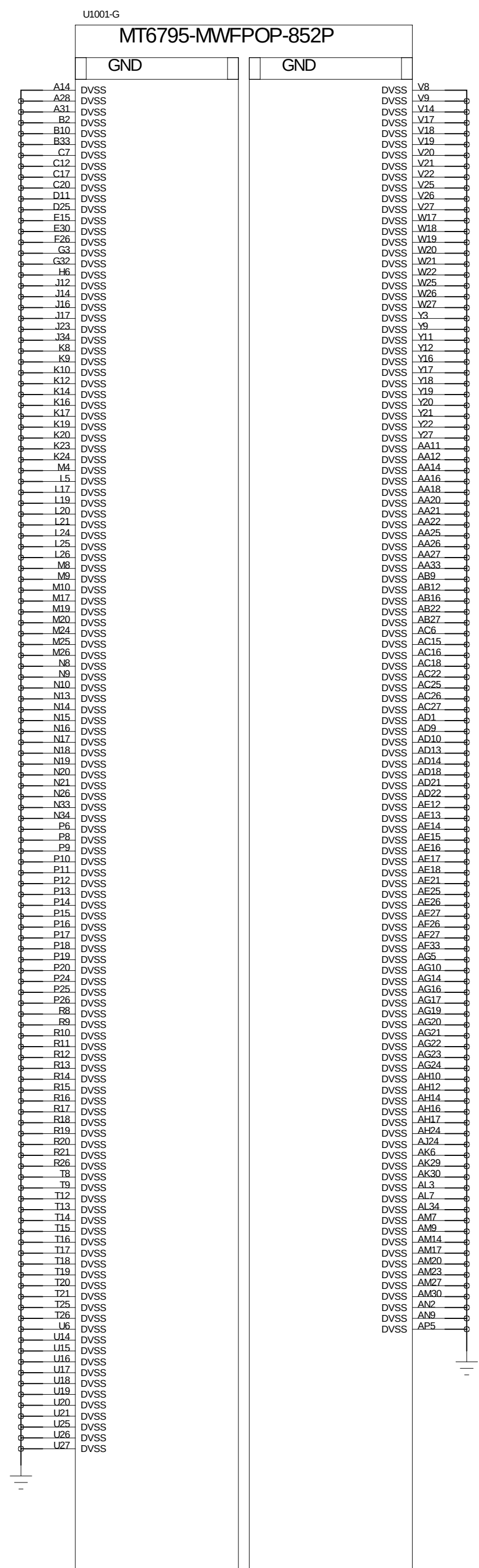
SHIELDING AND MARK

SH1001	SH1002	SHC1001	SHC1002	SHC1003	SHC1004	SHC1005	MARK1001	MARK1002
SH-4IN1-H951	SH-H951	SH-H951	SH-H951	SH-H951	SH-H951	SH-H951	SH-H951	SH-H951
SH1003	SH1004	LG1001	SN1001	WPL1001	H1	H2	H3	H4
SH-BB-H951	SH-RF-H951	SH-INF-H951	SH-6.4H	SH-6.4H	SH-6.4H	SH-6.4H	SH-6.4H	SH-6.4H

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 10_BB_POWER_1		VERSION: V1.1	SHEET: 3 OF 22
CHECKED	< >	DATED	< >				

BB_POWER_2

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Schematic design notice of "11_BB_POWER" page.

Note 11-1: AVDD28_DAC (Y33 ball) must be powered by "VTCXO 1 PMU".

Note 11-2: Connect VCC18IO_CM_R (E26 ball) to "VIO18_P_{MI}U" since I2C-2 I/O power is powered by VCC18IO_CM_R.

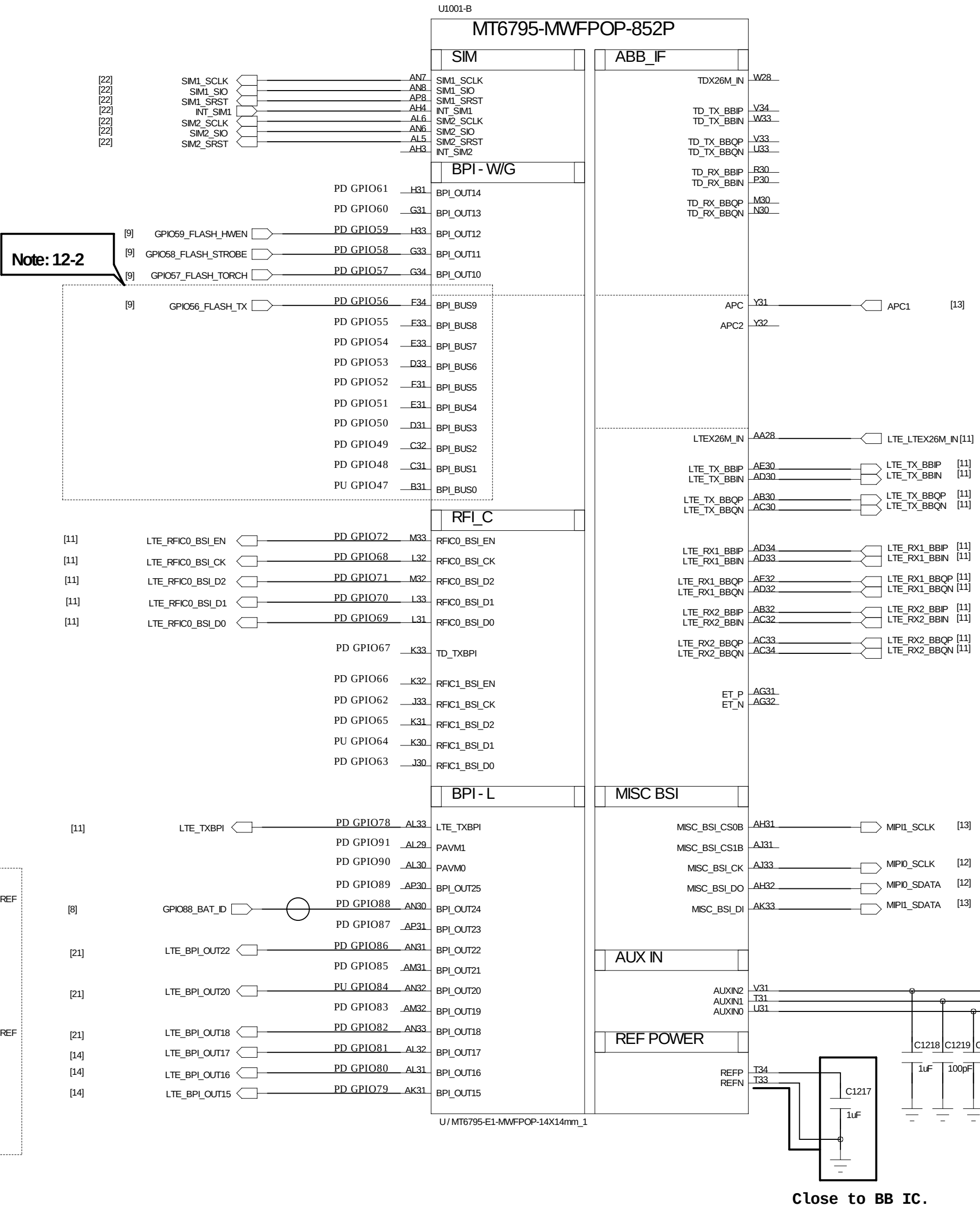
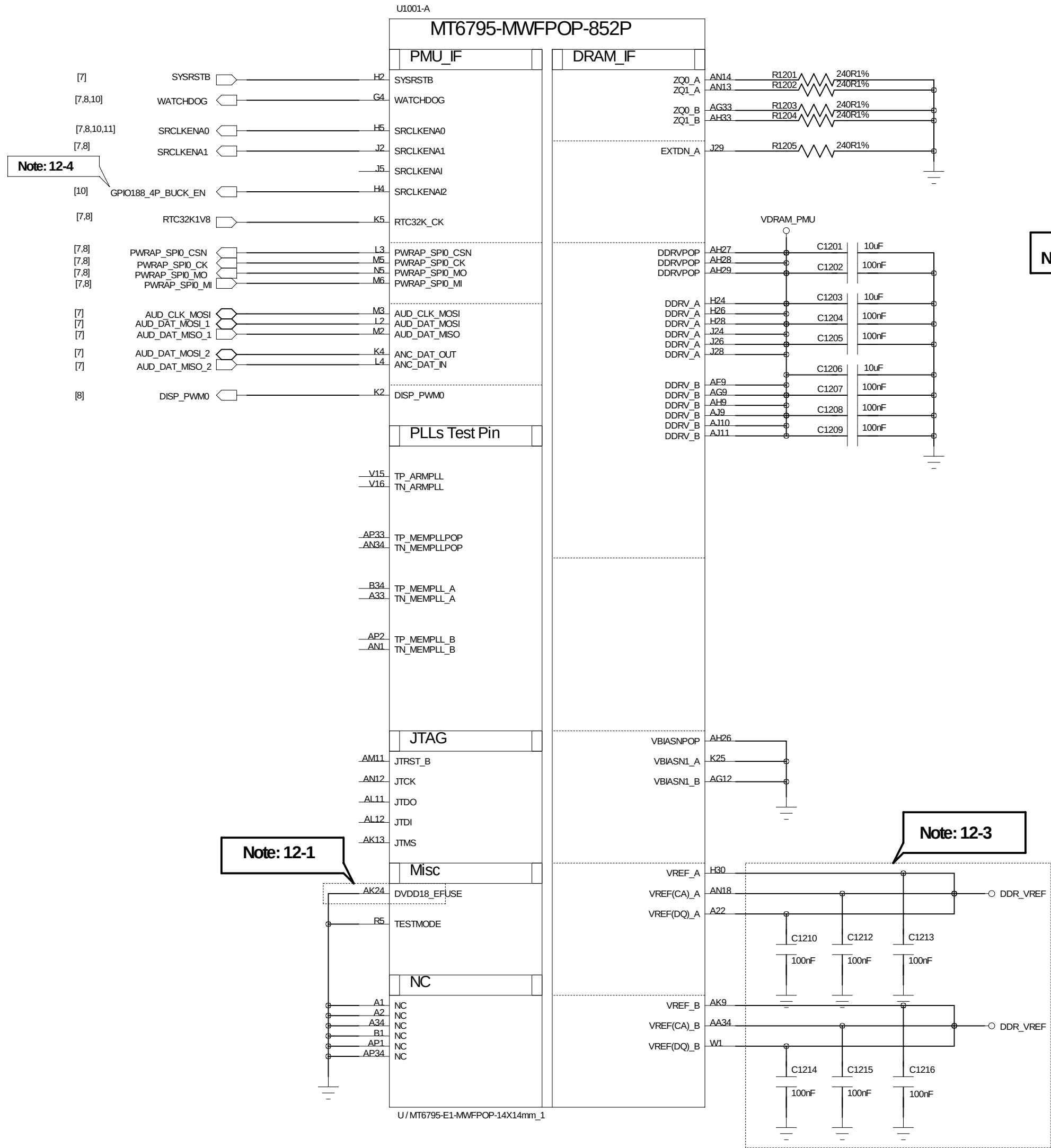
Note 11-3: Connect AVDD33_USB_P1 (B4 ball) to "VSIM1_PMU" for IC-USB / Samrt card application.

Connect AVDD33_USB_P1 (B4 ball) to "VUSB33 PMU" for USB application.

Note 11-4: Reserve 0.1uF decoupling cap in VCC18IO DPI for MHL.

COMPANY: U / M76795-E1-AMFPOP-14X14mm Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 11_BB_POWER_2		VERSION: V1.1	SHEET: 4 OF 22
CHECKED	< >	DATED	< >				

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



Schematic design notice of "12_BB_11" page.

Note 12-1: Apply 1.8V to DVDD18_EFUSE (AK24) for eFuse programming.

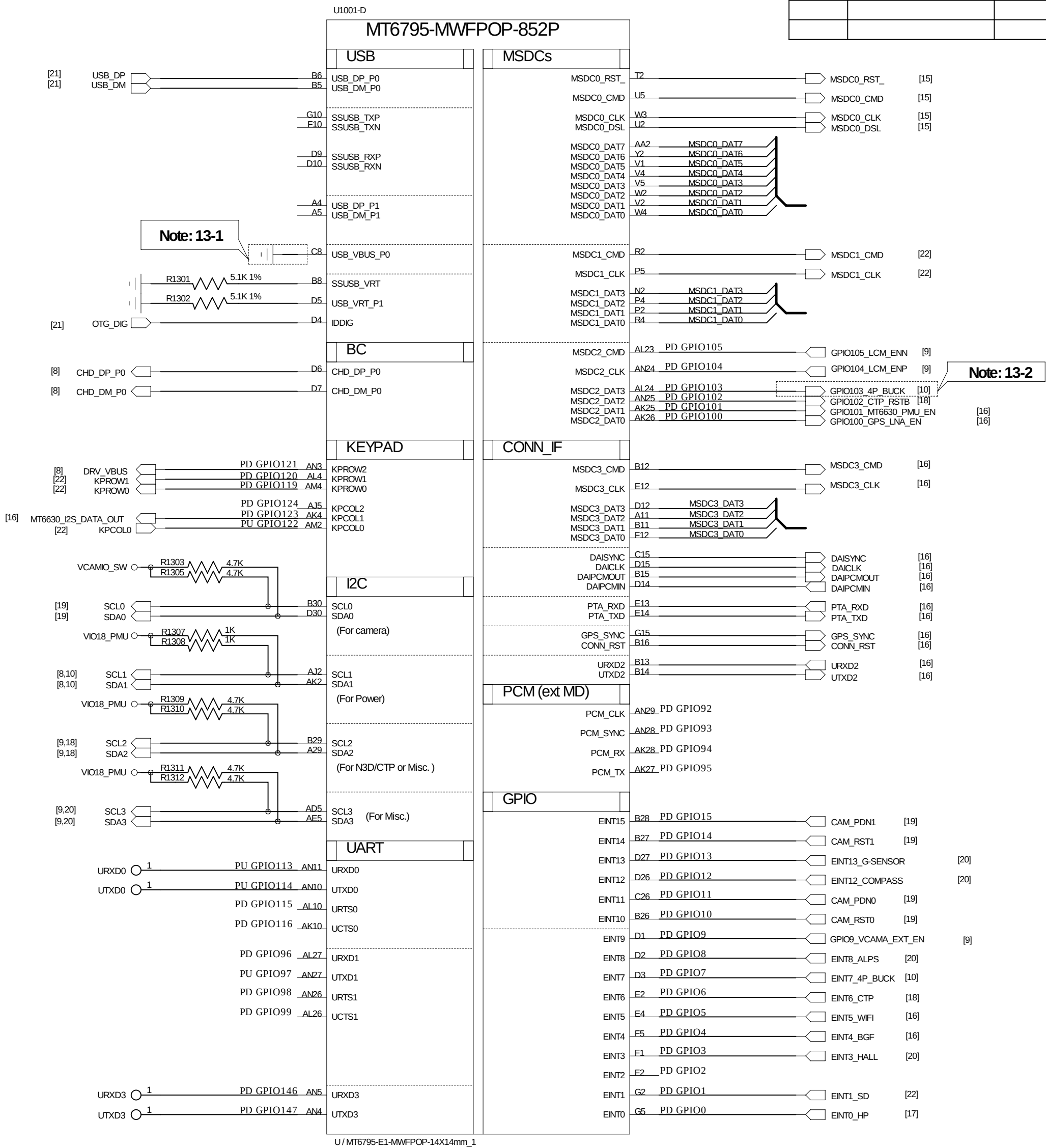
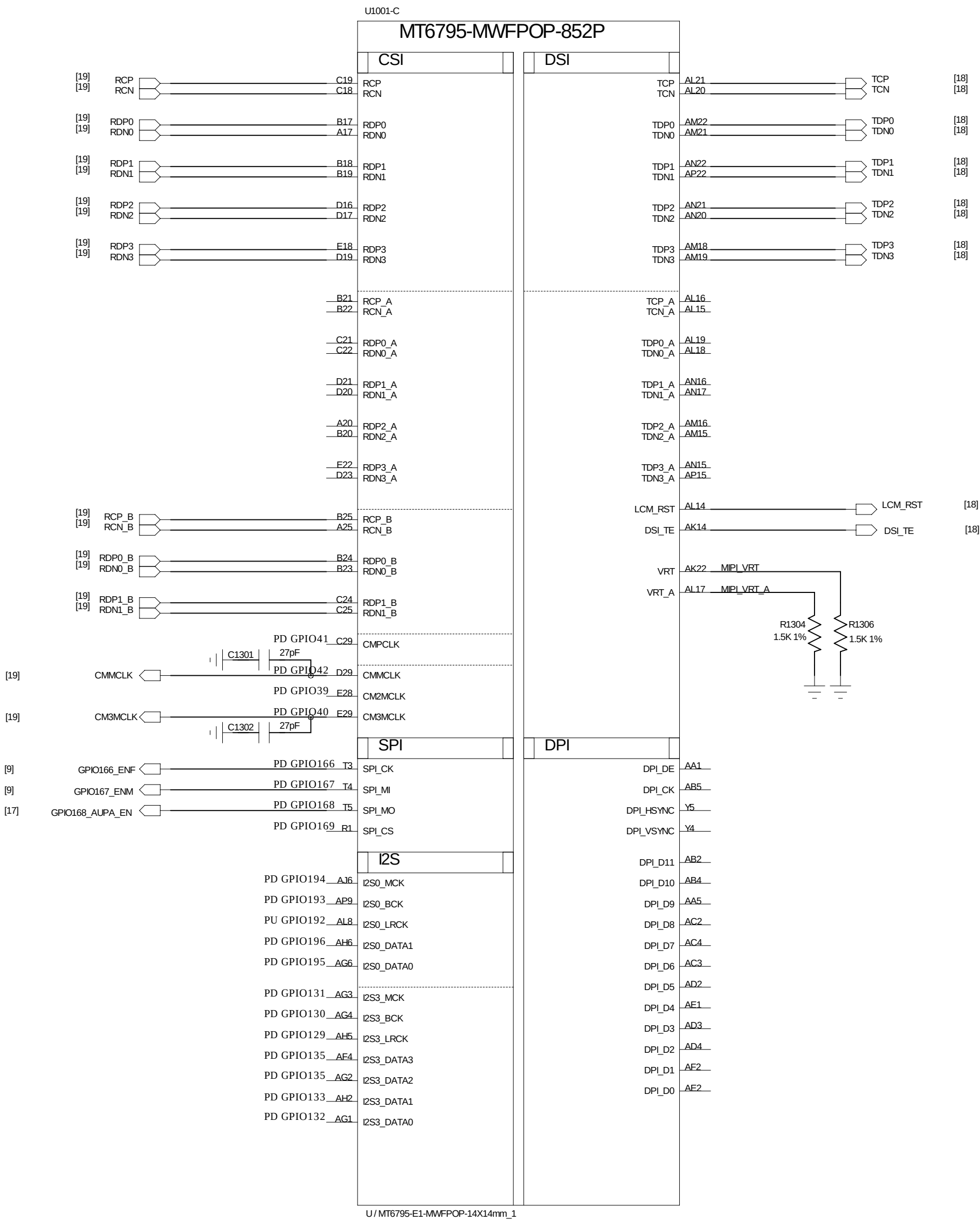
Note 12-2: The BPI_BUS0~BPI_BUS9 are capable of 2.8V I/O operation.

Note 12-3: The de-coupling cap. of DRAM VREF have to be placed as close to BB as possible.

Note 12-4: SRCLKENA12 features watch dog reset output to reset 4-phase buck.

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 12_BB_1		VERSION: V1.1	SHEET: 5 OF 22
CHECKED	< >	DATED	< >				

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



Schematic design notice of "13_BB_2" page.

Note 13-1: Connect USB_VBUS_P0 (C8 ball) pin to GND since USB OTG "B-Valid" detection has implemented by MT6332's ADC.

Note 13-2: GPIO103 is dedicated for DA9210 4-phase buck control.

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952	Modified Date: 2015-9-1
DRAWN	ZY/TS	DATED	20140512	TITLE: 13_BB_2	VERSION: V1.1 SHEET: 6 OF 22
CHECKED	< >	DATED	< >		

POWER_MT6331

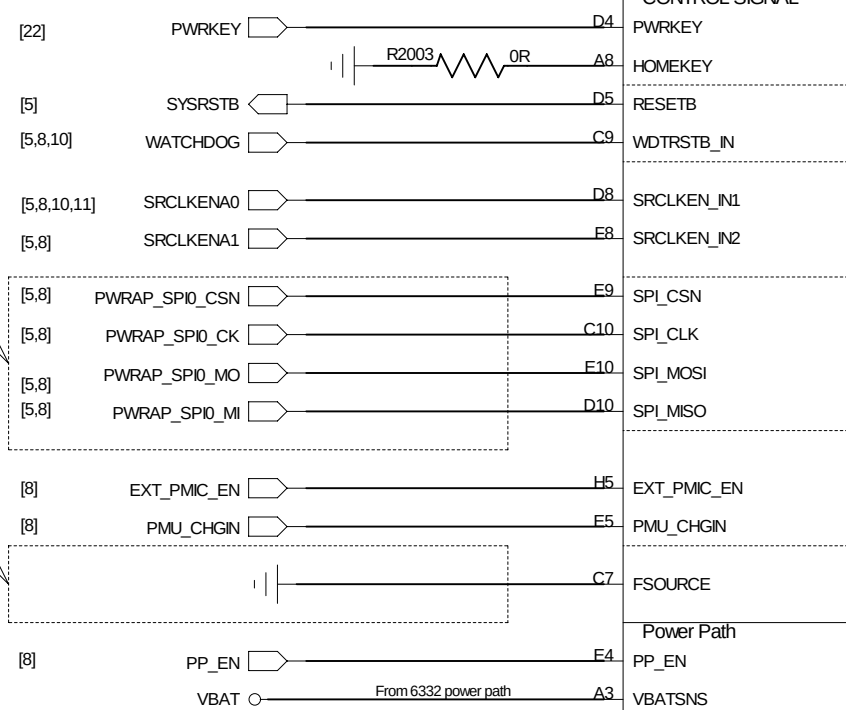
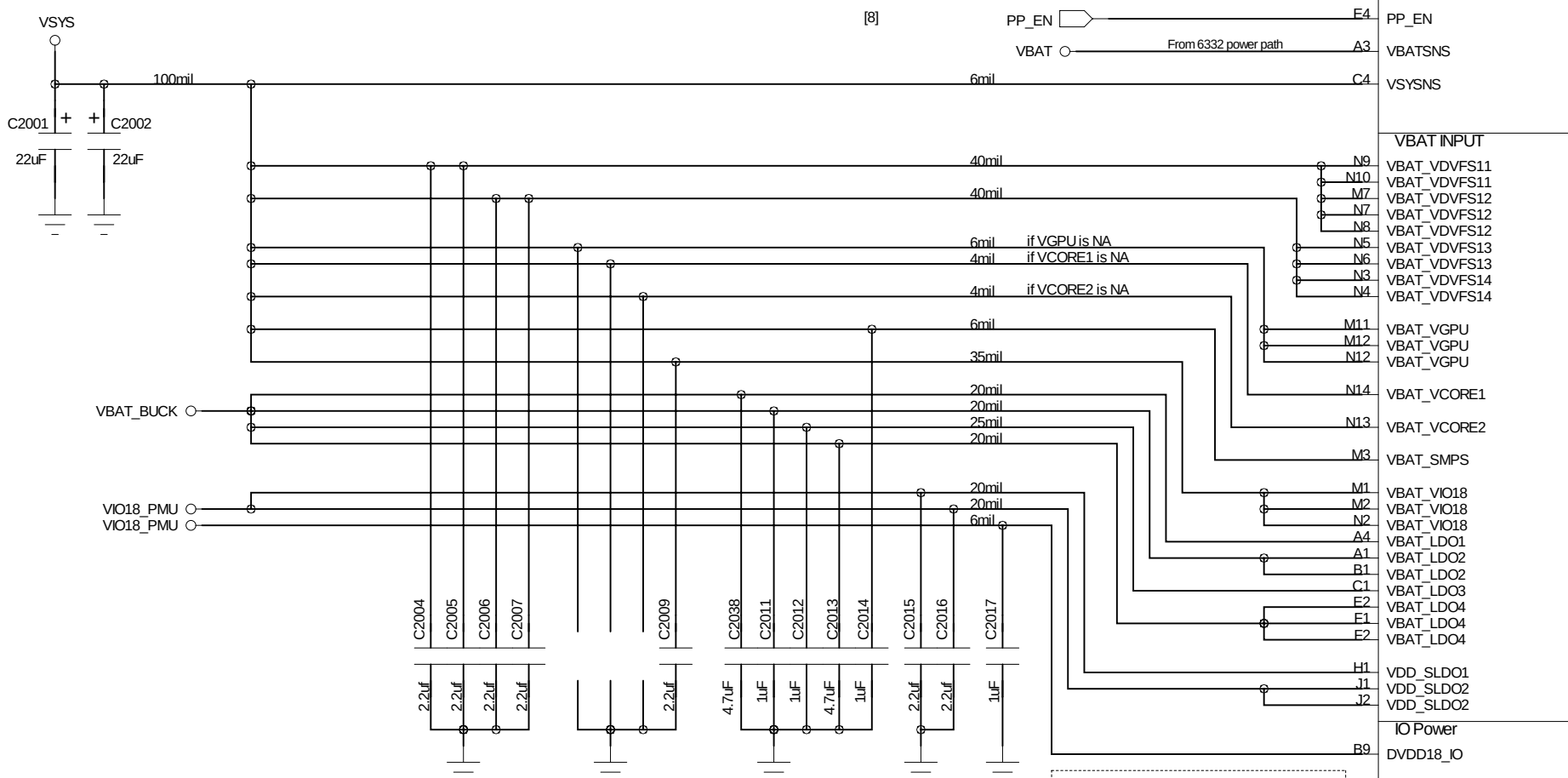
Schematic design notice of "20_POWER_MT6331" page.

Note 20-1: External pull resistor in PMU SPI interface is not allowed.

Note 20-2: This PIN should be connected to ground for normal operation.

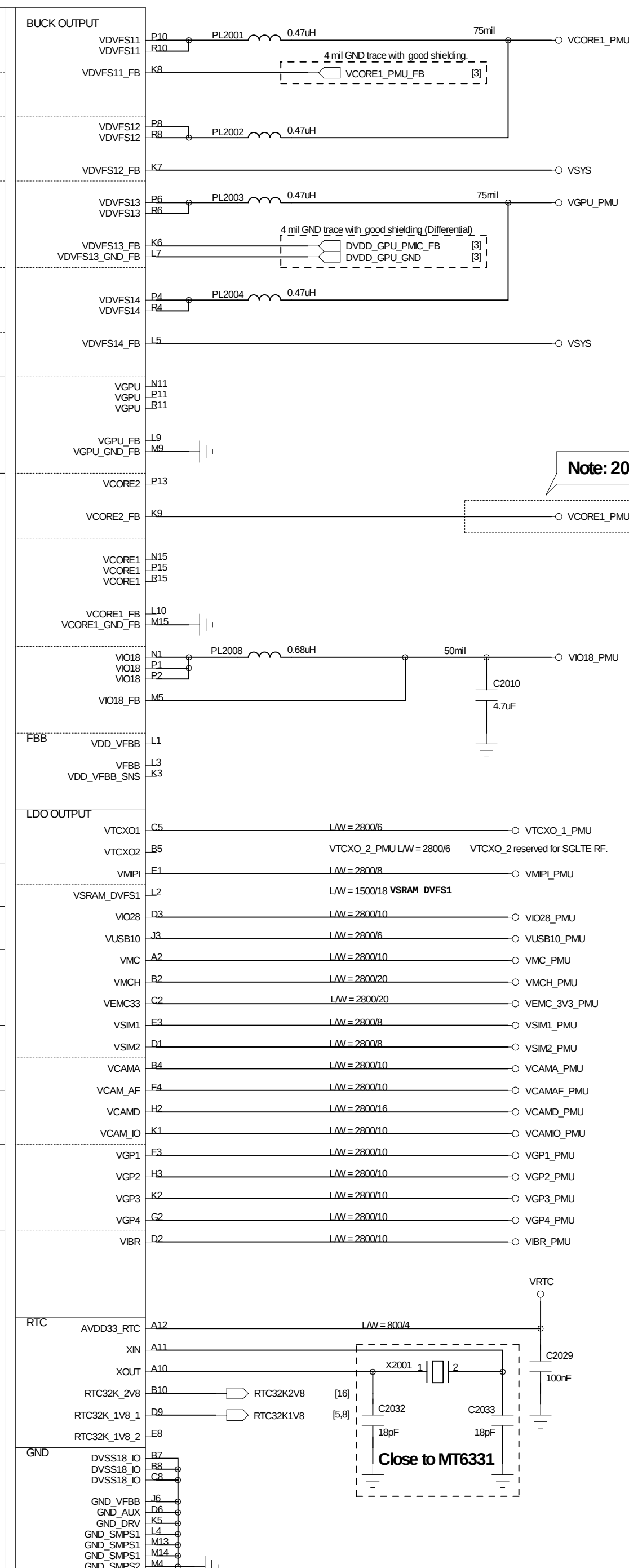
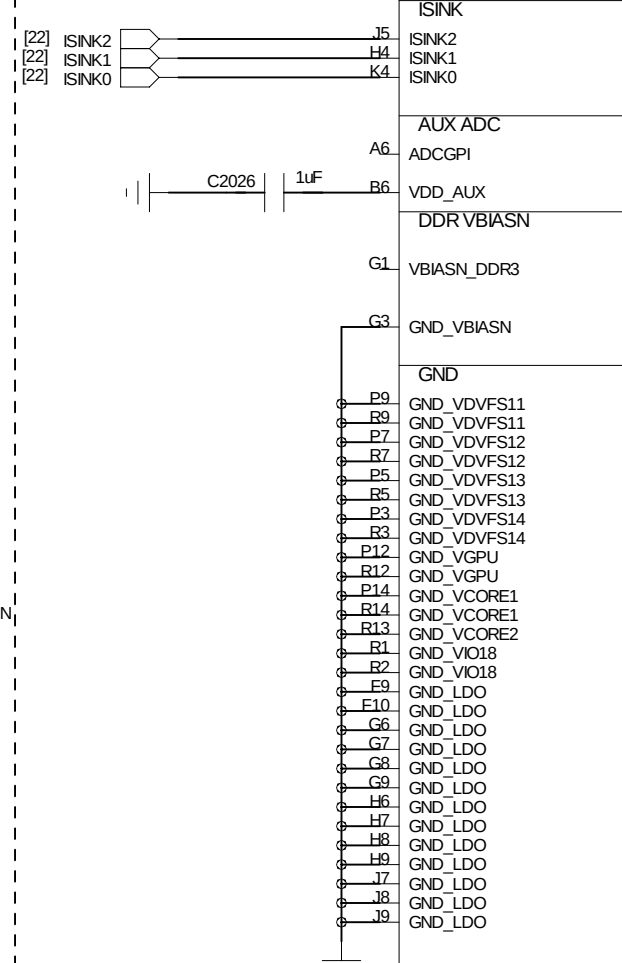
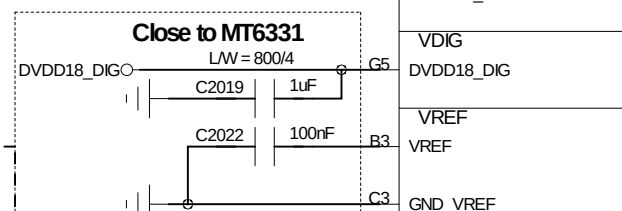
Note 20-3: Connect MT6331's K9 pin to "VCORE1_PMU" when VCORE2 is not used.

Note 20-4: Layout trace from MT6331 ball D12 AU_REFN to Audio jack GND must surround shield with GND



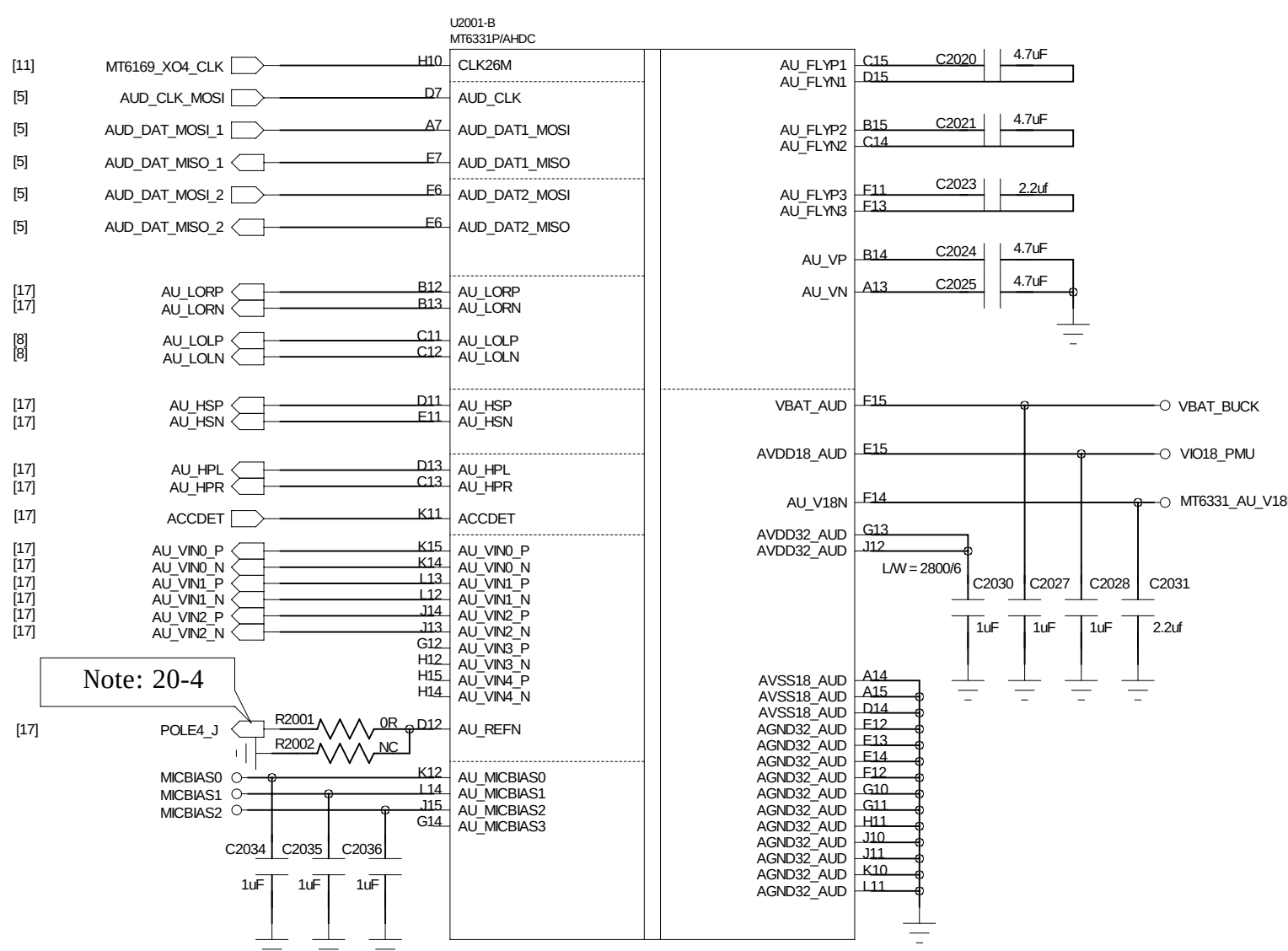
Note: 20-1

Note: 20-2



Note: 20-3

Audio Part



Note: 20-4

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

PowerType	Power	Available Vout	Default V	Max current
BUCK Current Regulator	VDVFS11	0.7V-1.31V	1.05V	2000mA
	VDVFS12			
	VDVFS13			
	VDVFS14			
		VCORE2	0.7V-1.31V	1.05V
	VIO18	1.8V	1.8V	2400mA
Analog LDO	VTCXO1	2.8V	2.8V	40mA
	VTCXO1	2.8V	2.8V	40mA
	AVDD32_AUD	3.2/3.0/2.8V	3.2V	80mA
	VAUXA32	3.2/3.0/2.8V	3.2V	20mA
	VCAMA	1.5/1.8/2.5/2.8V	2.8V	250mA
Digital LDO	VIO28	2.8V	2.8V	200mA
	VCAM_AF	1.2/1.3/1.5/1.8V 2.0/2.8/3.0/3.3V	2.8V	200mA
	VMC	1.8/3.3V	3.3V	200mA
	VMCH	3.0/3.3V	3.3V	800mA
	VEMC33	3.0/3.3V	3.3V	800mA
	VSIM1	1.7/1.8/1.86V 2.7/6.3/0.3.1V	1.8V	100mA
	VSIM2	1.7/1.8/1.86V 2.7/6.3/0.3.1V	1.8V	100mA
	VMPI	1.2/1.3/1.5/1.8V 2.0/2.8/3.0/3.3V	1.8V	150mA
	VMBR	1.2/1.3/1.5/1.8V 2.0/2.8/3.0/3.3V	2.8V	200mA
	VCAMD	0.9/1.0/1.1V 1.2/2.1/3.1.5V	1.3V	500mA
	VCAM_IO	1.2/1.3/1.5/1.8V	1.8V	200mA
	VUSB10	1.0/1.05/1.1V 1.15/1.2/1.25/1.3V	1.0V	60mA
	VSRAM_DVFS1	1.0125V	1.0125V	600mA
	VGP1	1.2/1.3/1.5/1.8V 2.0/2.8/3.0/3.3V	2.8V	200mA
	VGP2	1.1/1.2/1.35/1.5V	1.2V	10mA
	VGP3	1.2/1.3/1.5/1.8V	1.8V	200mA
	VGP4	1.6/1.7/1.8V 1.9/2.0/2.1/2.2V	2.0V	200mA
VRTC	2.8V	2.8V	2mA	

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd

MODEL: H952

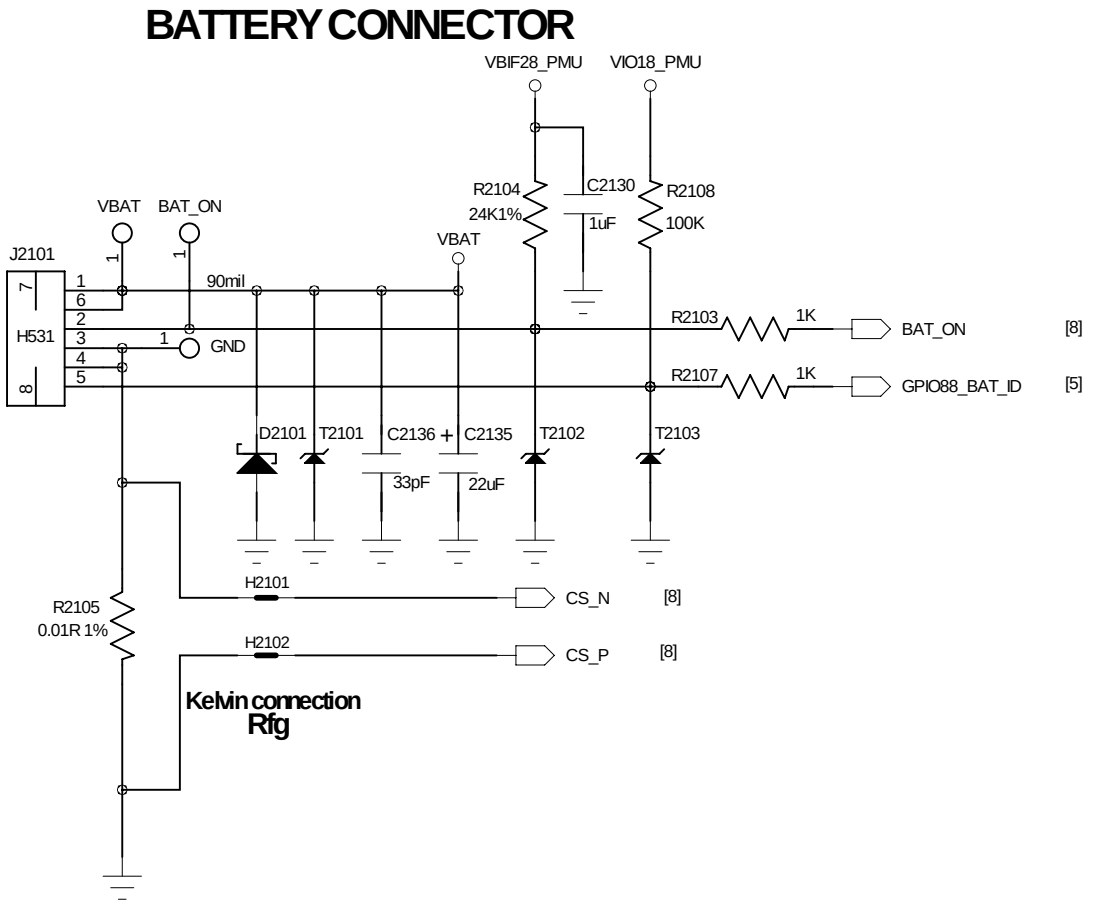
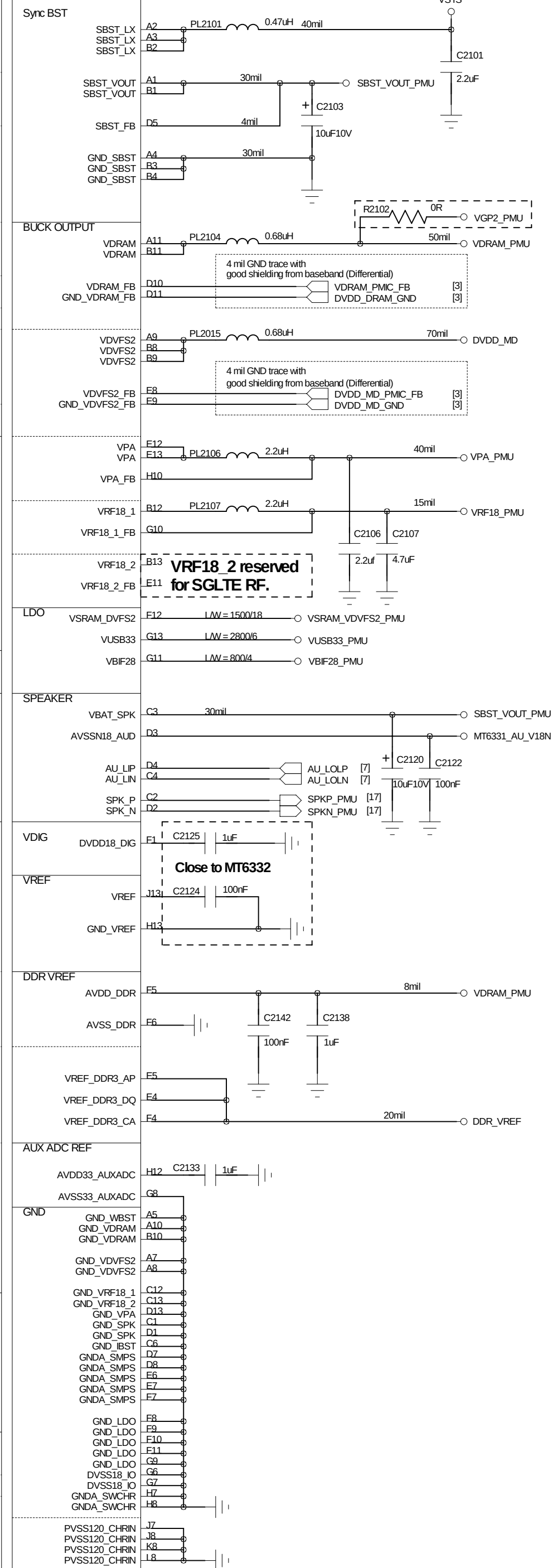
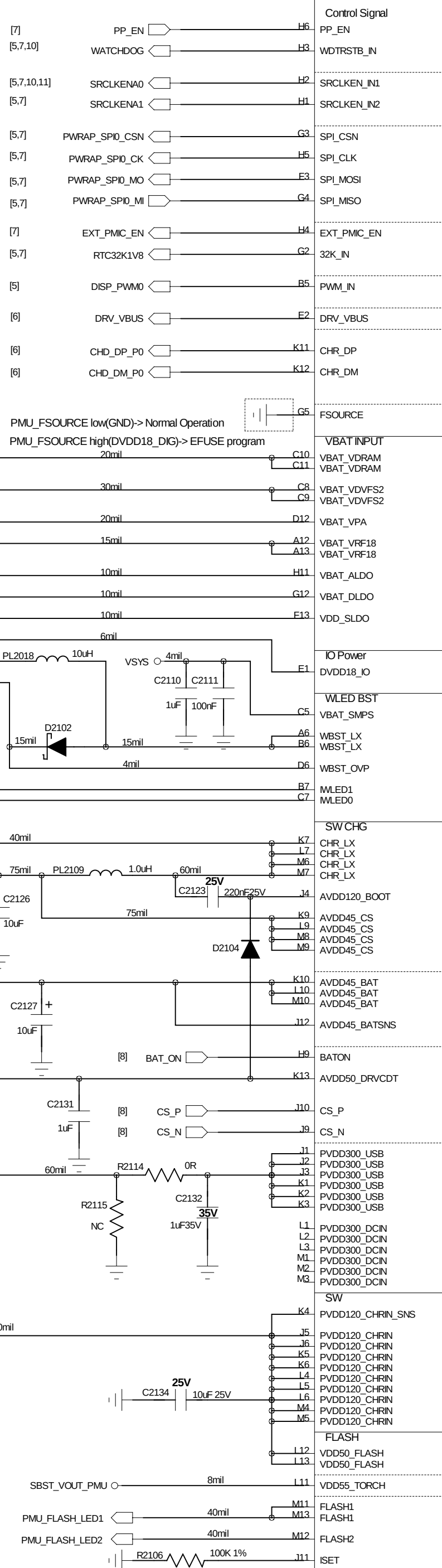
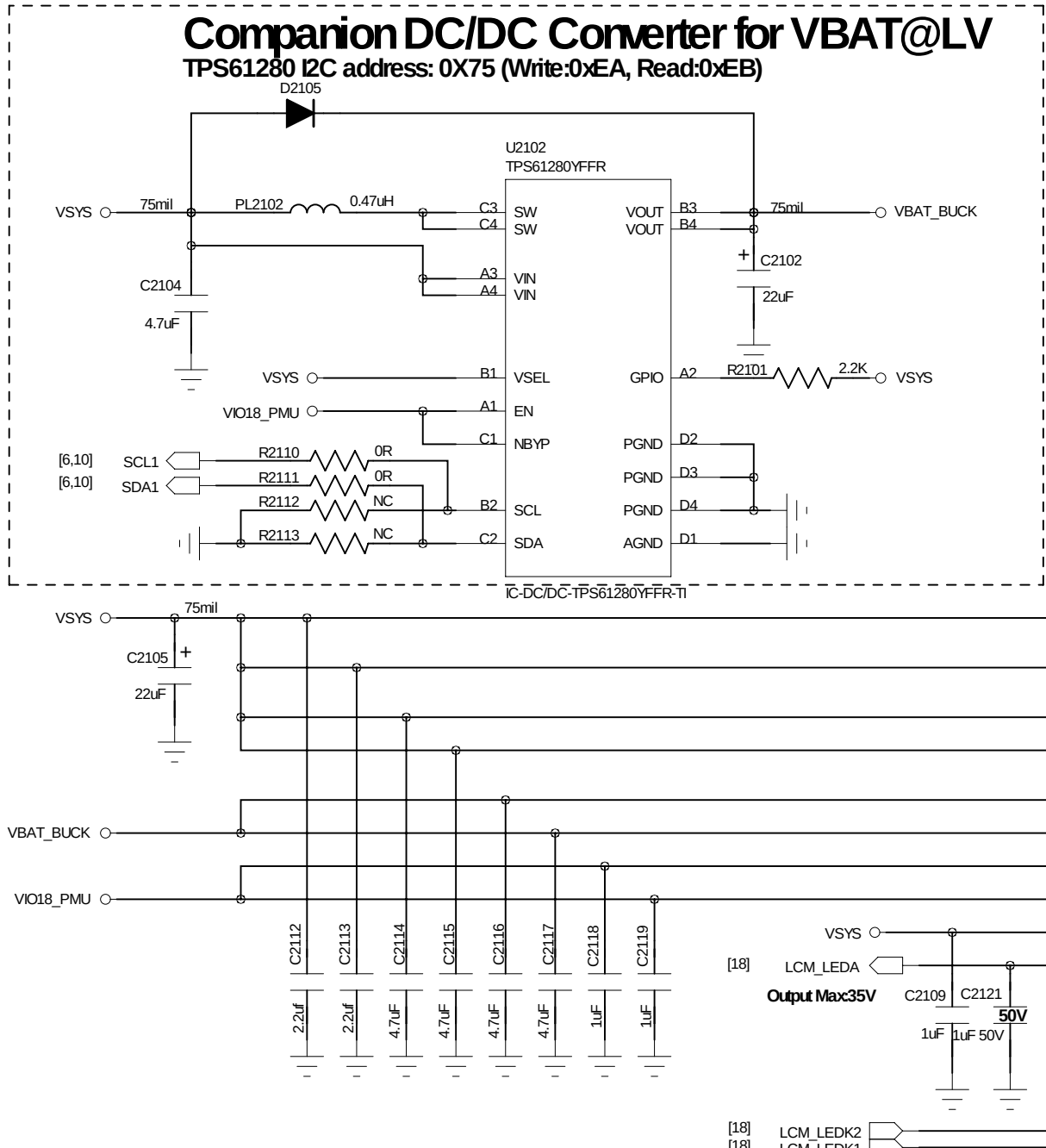
Modified Date: 2015-9-1

DRAWN	ZY/TS	DATED	20140512
CHECKED	< >	DATED	< >

TITLE: 20_POWER_MT6331

VERSION: V1.1	SHEET: 7 OF 22
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POWER_MT6332



REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

PowerType	Power	Available Vout	Default V	Max current
BUCK Converter	VDVFS2	0.7V-1.31V	1.0V	1400mA
	VDRAM	1.1V/1.2V	By trapping	2000mA
	VRF18_1	1.825V	1.825V	450mA
	VRF18_2	1.825V	1.825V	450mA
	VPA	0.5V-3.4V	0.5V	800mA

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952	Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 21_POWER_MT6332	VERSION: V1.1	SHEET: 8 OF 22
CHECKED	< >	DATED	< >			

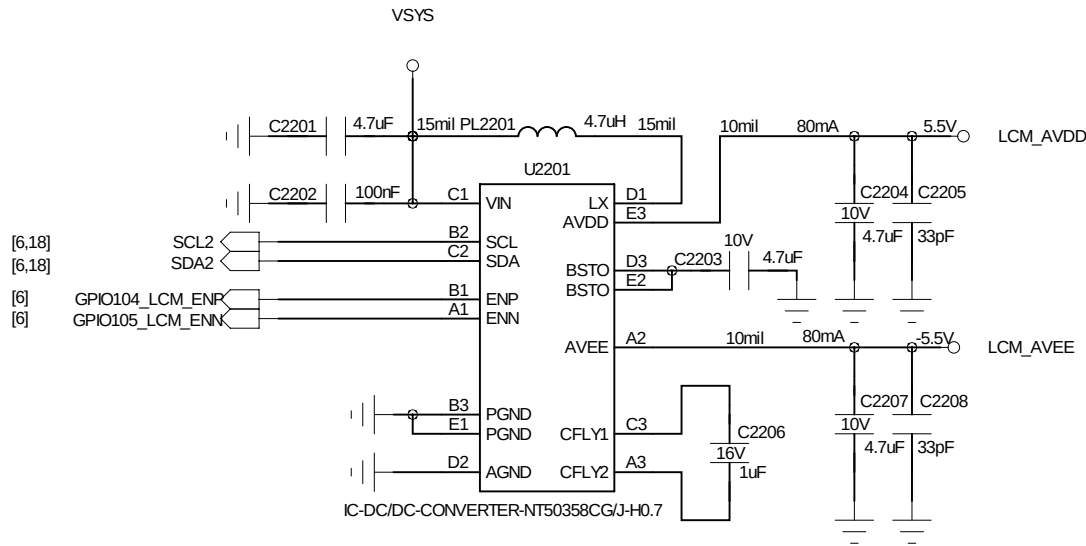
POWER_THIRD-PART

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

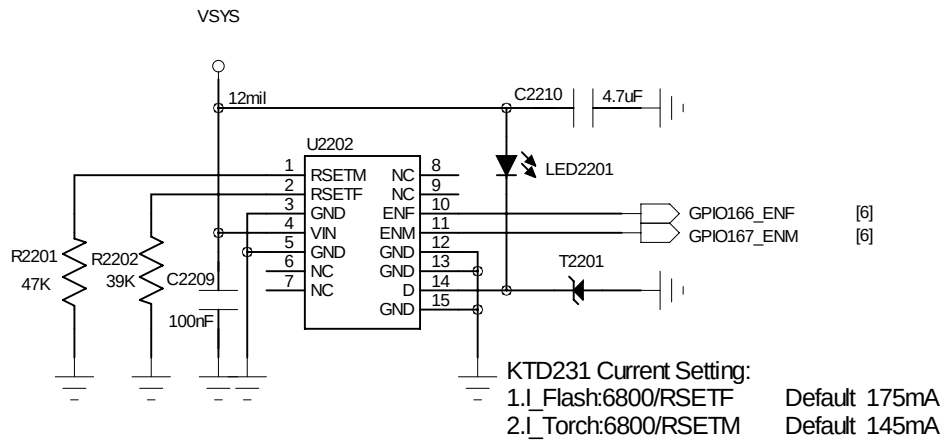
For 5.46" FHD

LCM BOOST

NT50358CG/J:I2C ADDRESS:0x7C(Write)/0x7D(Read)

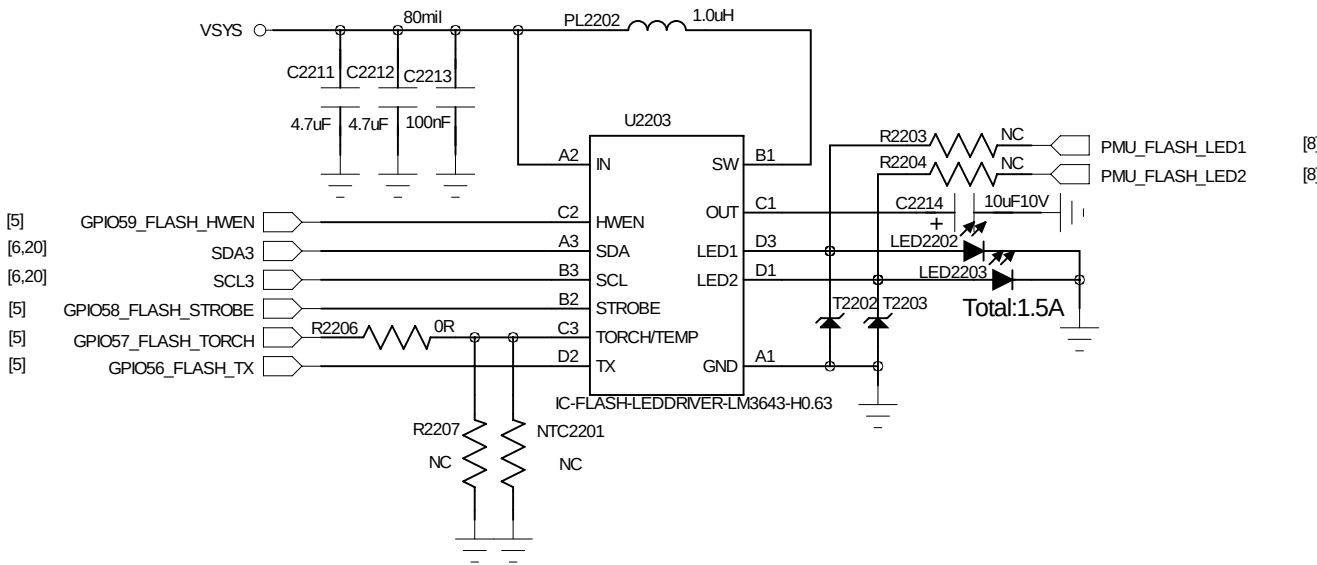


FRONT CAMERA LED

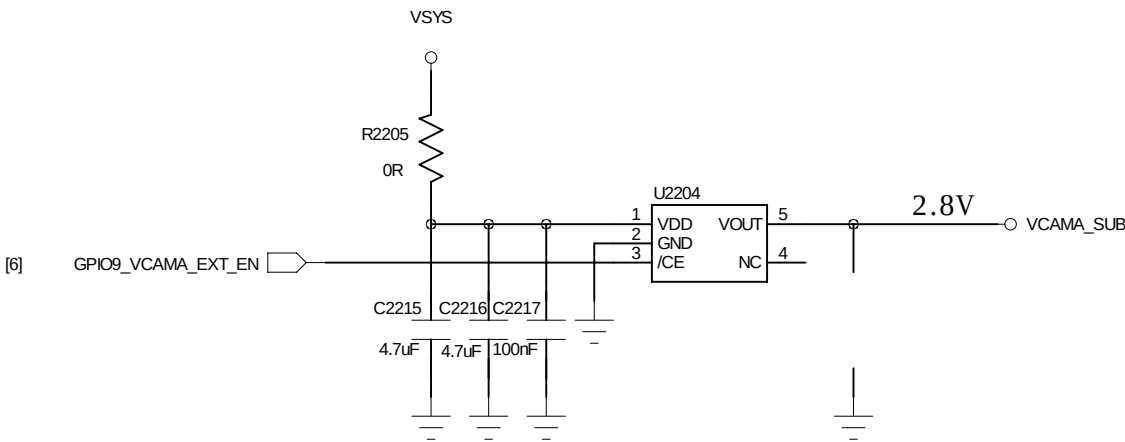


CAMERA FLASH LED

LM3643YFFR:I2C ADDRESS:0xC6(Write)/0xC7(Read)



2.8V_VCAMA_EXT for SUB CAM



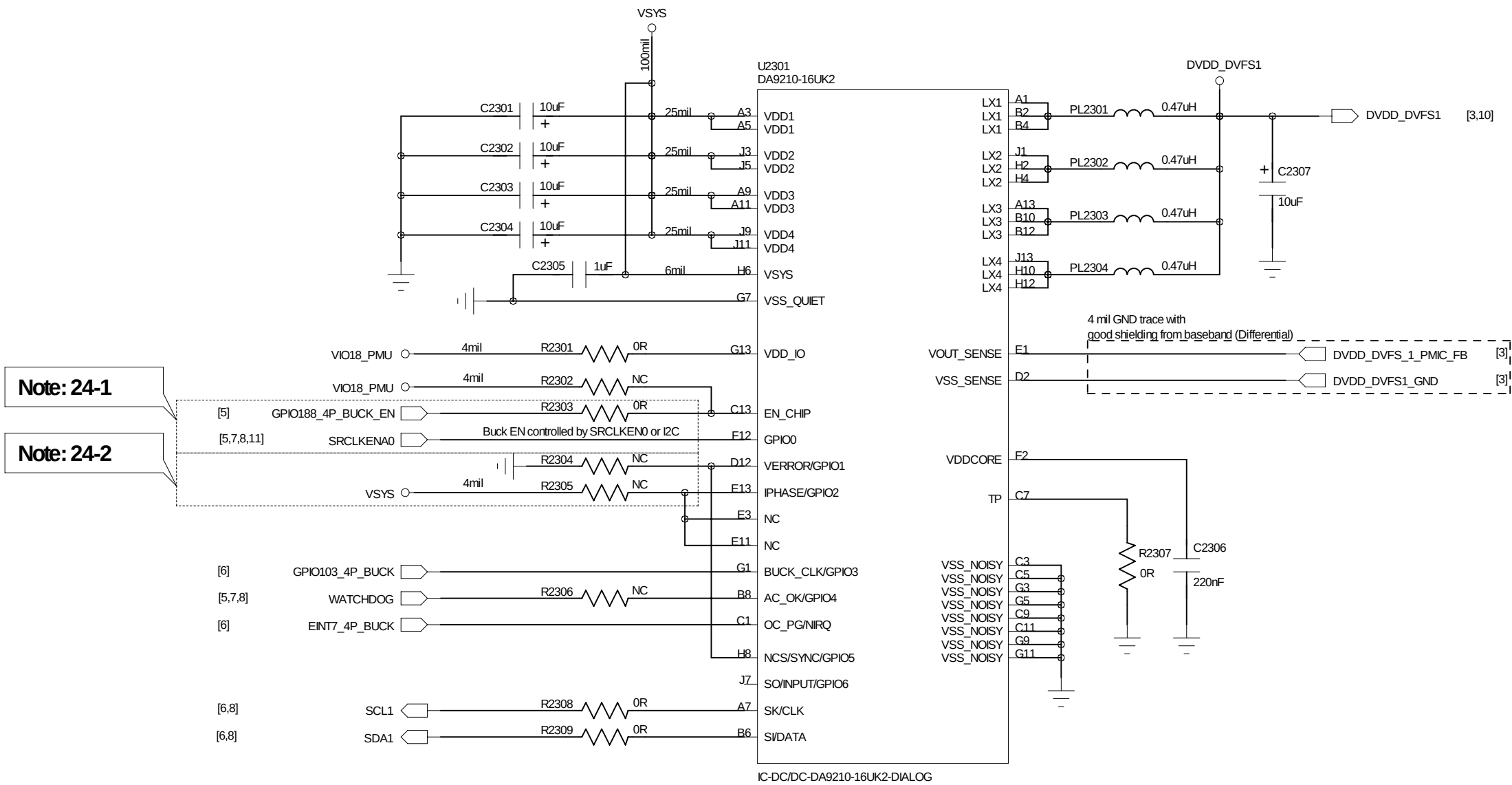
COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 22_POWER_THIRD-PARTY		VERSION: V1.1	SHEET: 9 OF 22
CHECKED	< >	DATED	< >				

POWER_EXT_4PHASE_BUCK

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

4-Phase Buck

4-Phase Buck I2C address: 0X68 (Write:0xD0, Read:0xD1)



Schematic design notice of "23_POWER_EXT_4PHASE_BUCK" page.

Note 24-1: DA9210's GPIO0 = SRCLKENA0 => Buck EN controlled by SRCLKEN0 or I2C.

Note 24-2: BOM option to select which 4-pahse buck be applied? DA9210-21UK2 or second source.

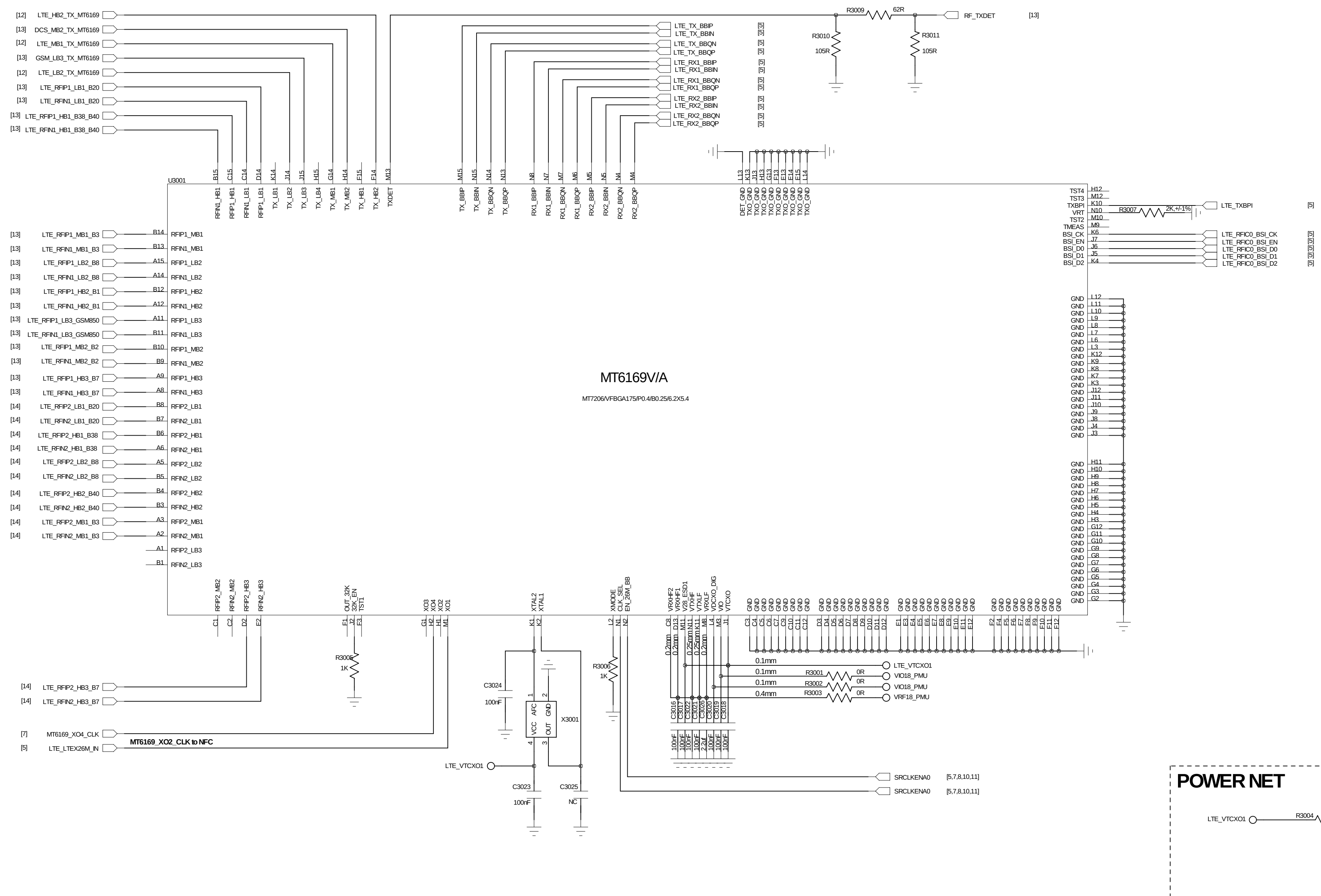
	R2302	R2303	R2304	R2305	R2306
DA9210-21UK2	N/M	0R	N/M	N/M	N/M
2nd source	10K	N/M	0R	0R	0R

Note 24-3: The part number of DA9210 must be specified to "DA9210-21UK2" for MT6795.

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952	Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 23_POWER_EXT_4PHASE_BUCK	VERSION: V1.1	SHEET: 10OF 22
CHECKED	< >	DATED	< >			

RF_MT6169_PIN_OUT

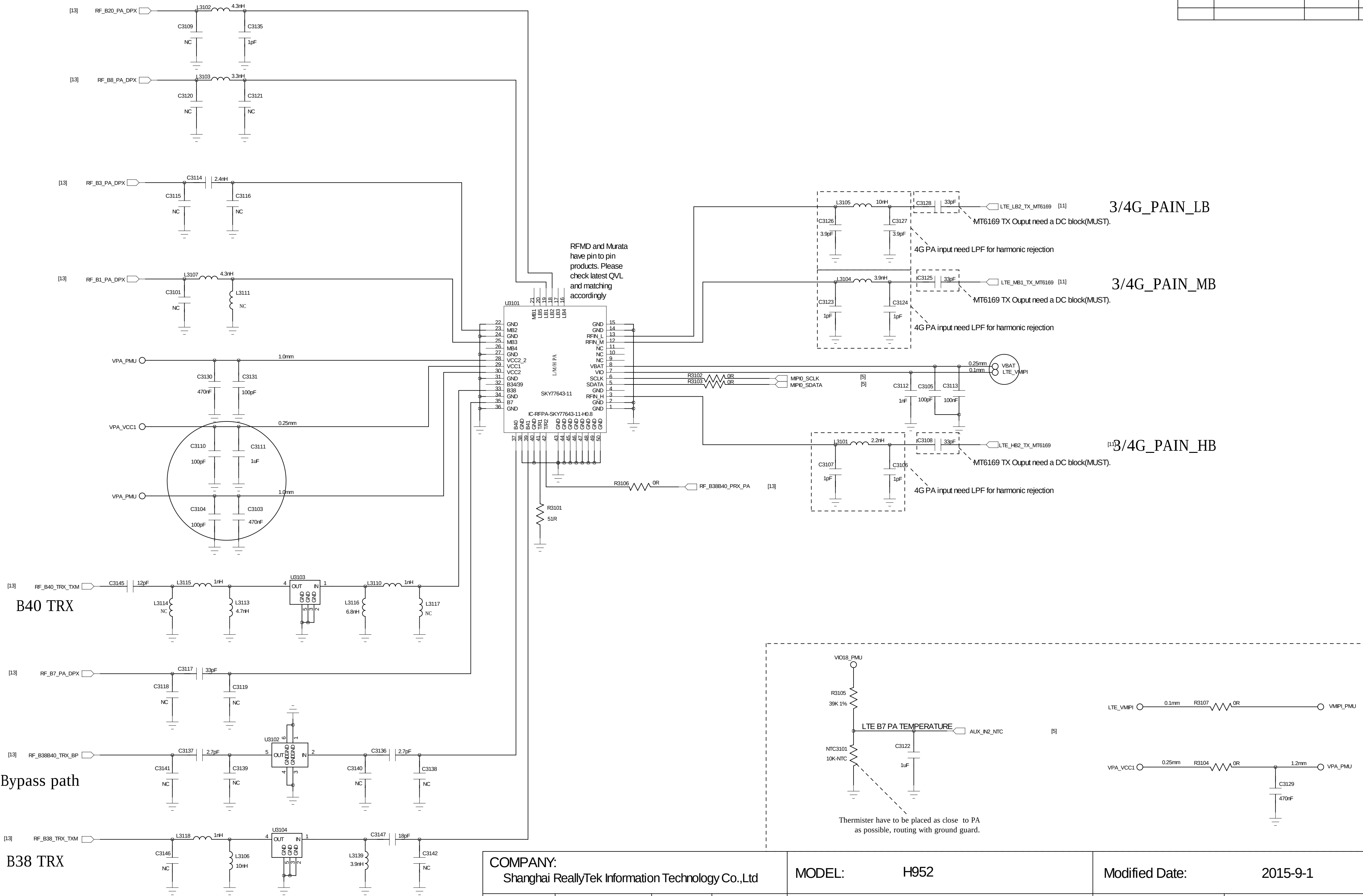
REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 30_RF_MT6169_PIN_OUT		VERSION: V1.1	SHEET: 11 OF 22
CHECKED	< >	DATED	< >				

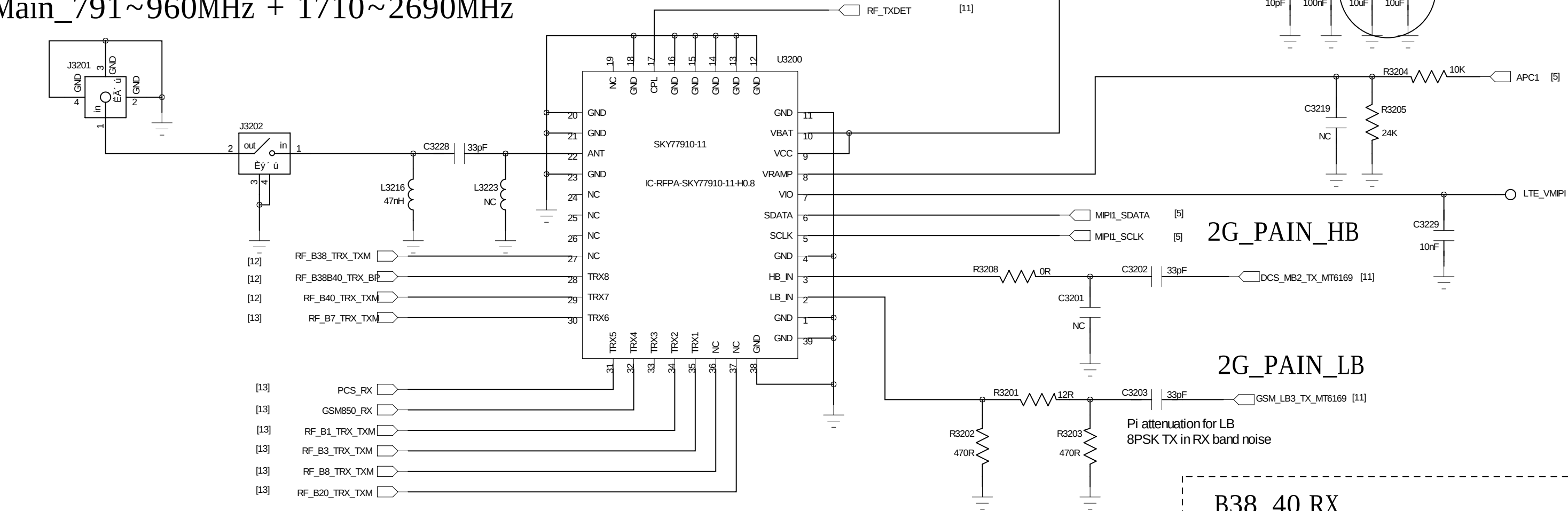
RF_MT6169_RF_TX

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



RF_MT6169_RF_PRX

ASM_Main_791~960MHz + 1710~2690MHz

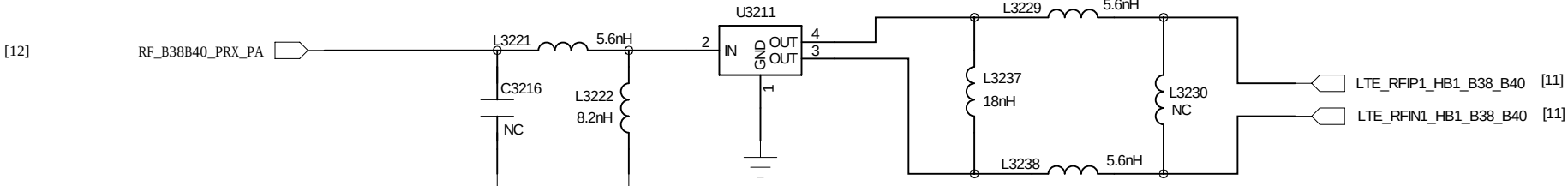


2G_PAIN_HB

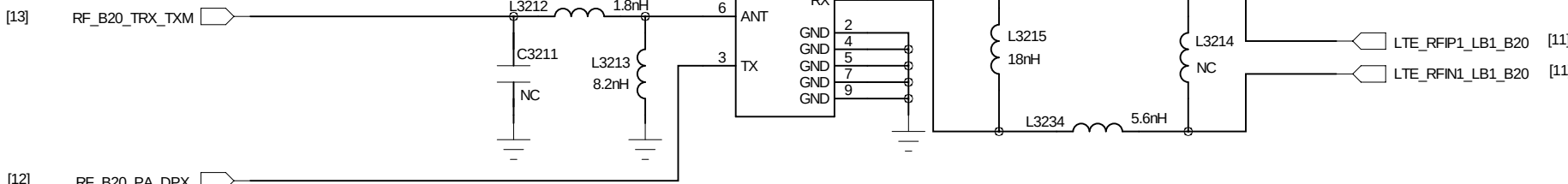
2G_PAIN_LB

Pi attenuation for LB
8PSK TX in RX band noise

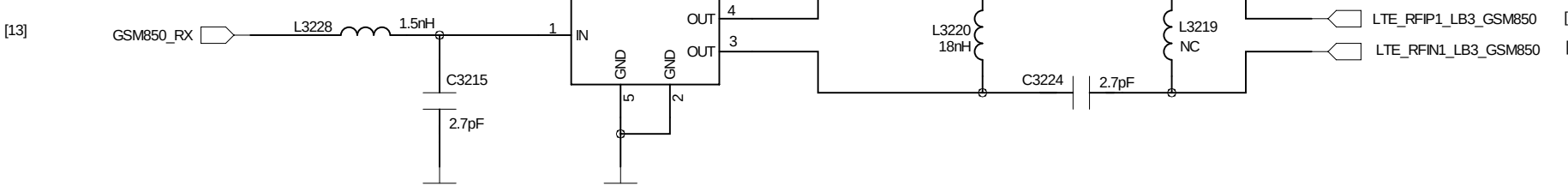
B38_40 RX



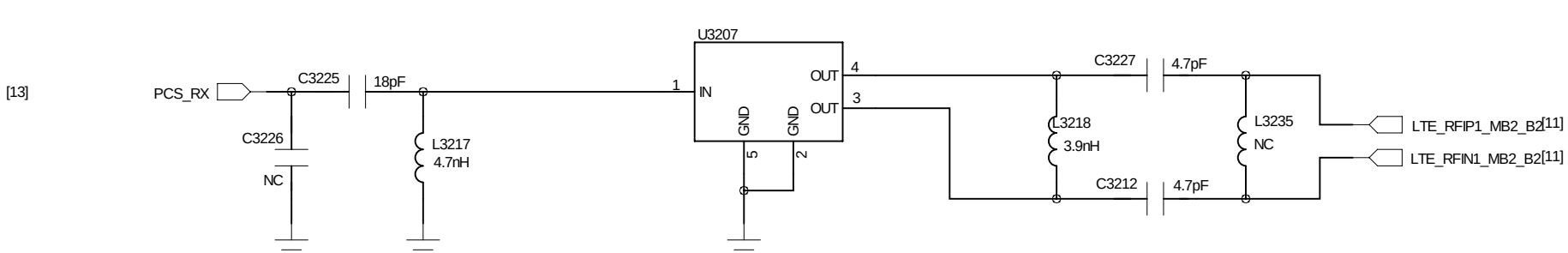
B20 TRX



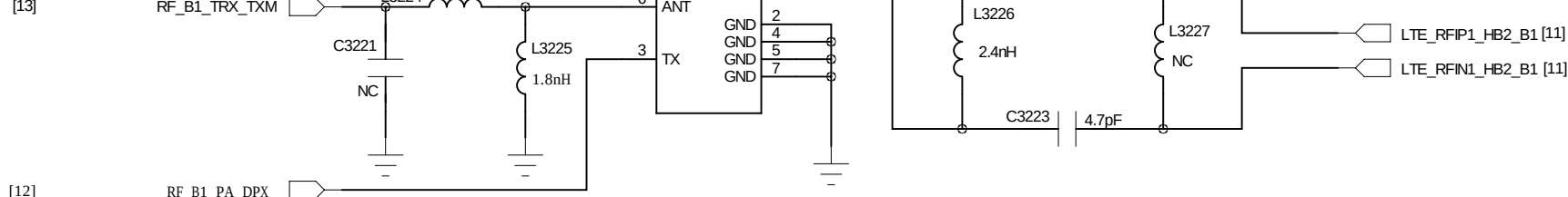
GSM850



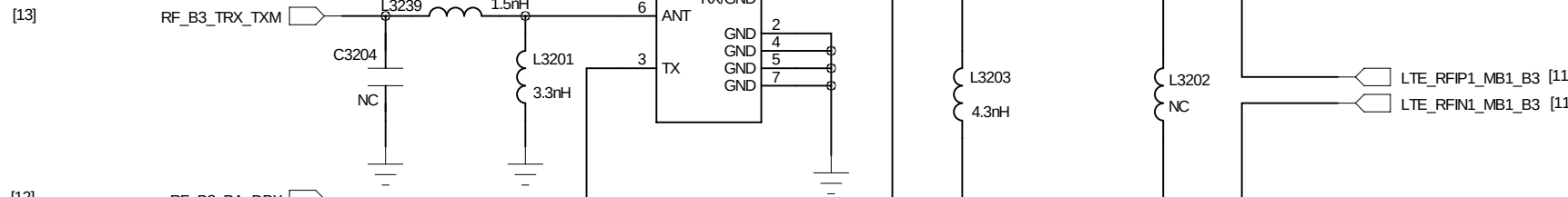
B2 RX



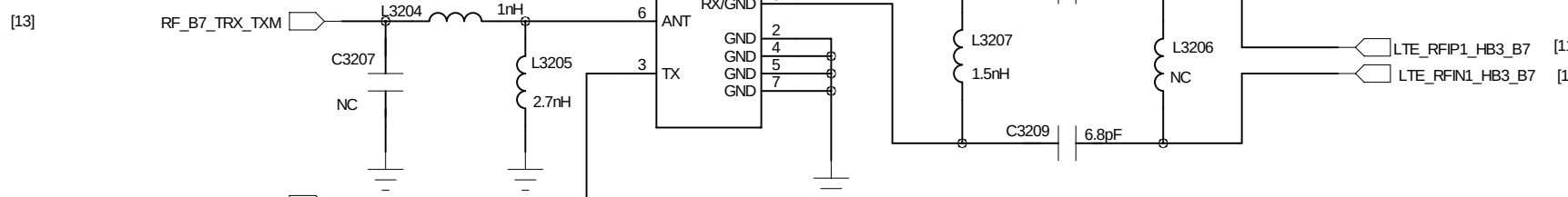
B1 TRX



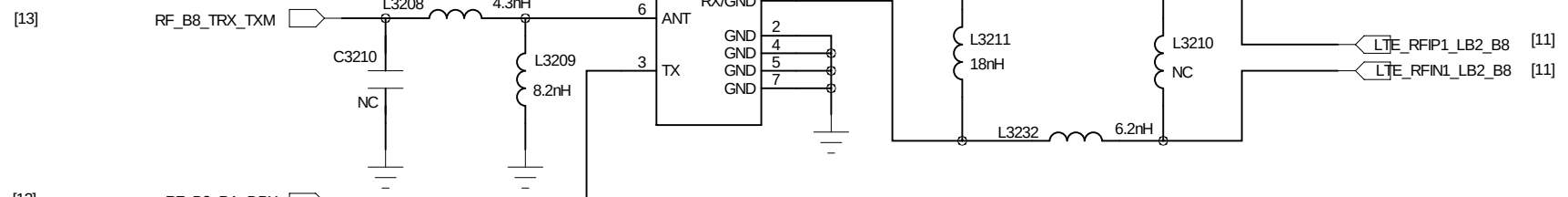
B3 TRX



B7 TRX



B8 TRX



COMPANY:
Shanghai ReallyTek Information Technology Co.,Ltd

MODEL: H952

Modified Date: 2015-9-1

DRAWN ZY/TS

DATED 20140512

TITLE: 32_RF_MT6169_RF_PRX

VERSION: V1.1

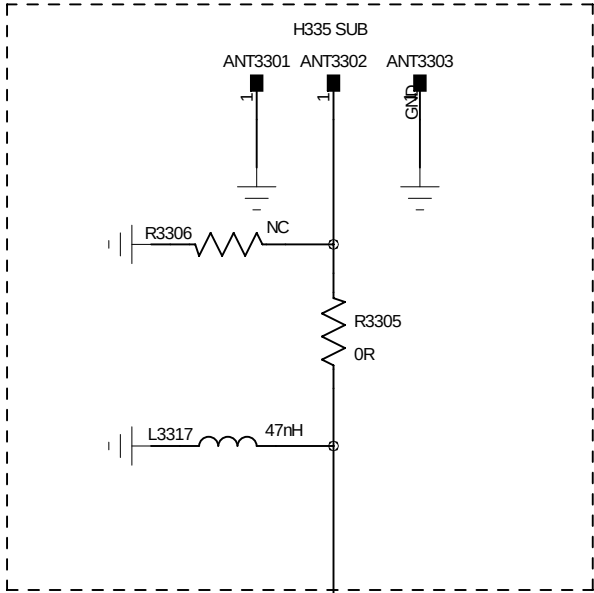
SHEET: 13OF 22

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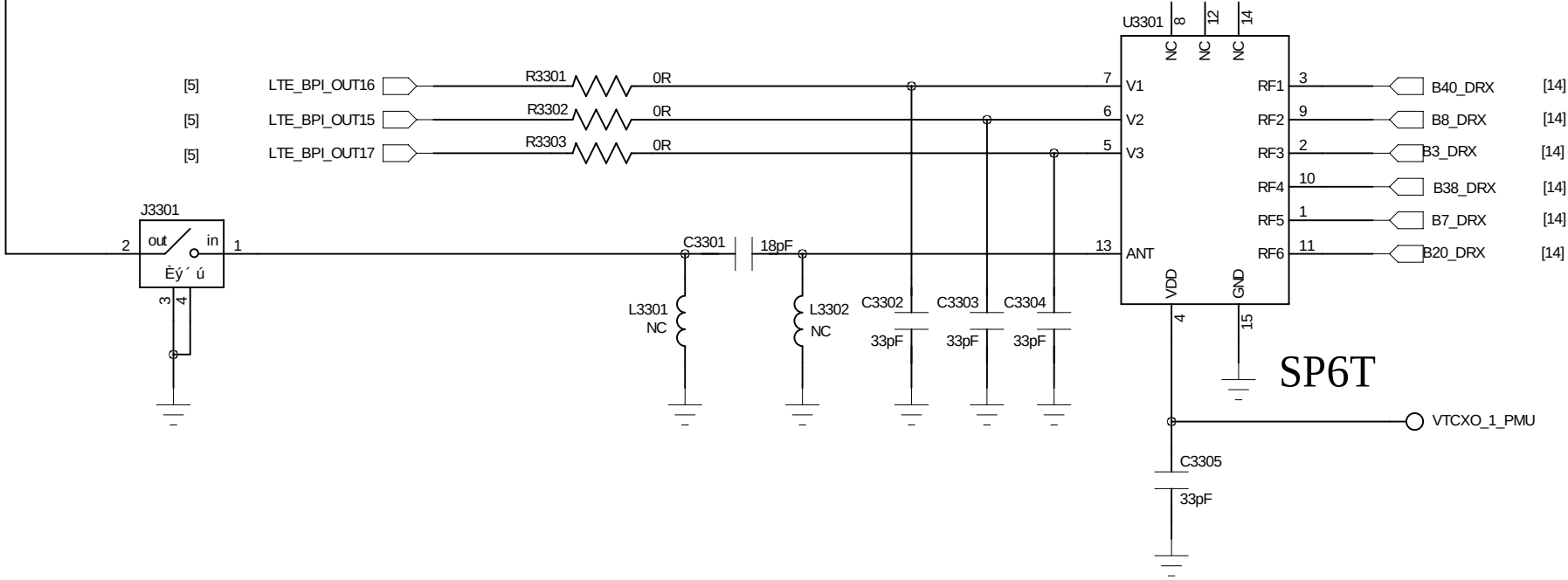
DATED < >

RF_MT6169_RF_DRX

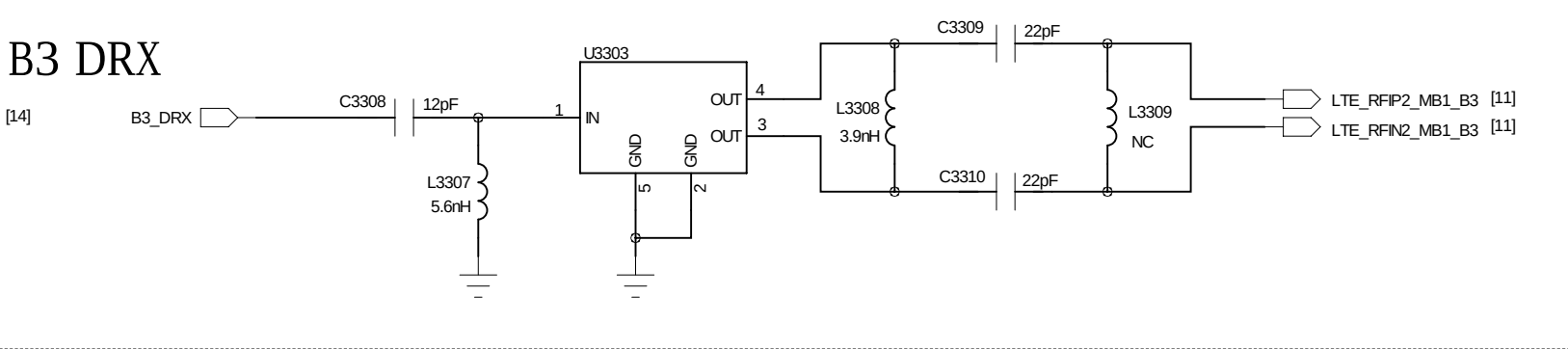
REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



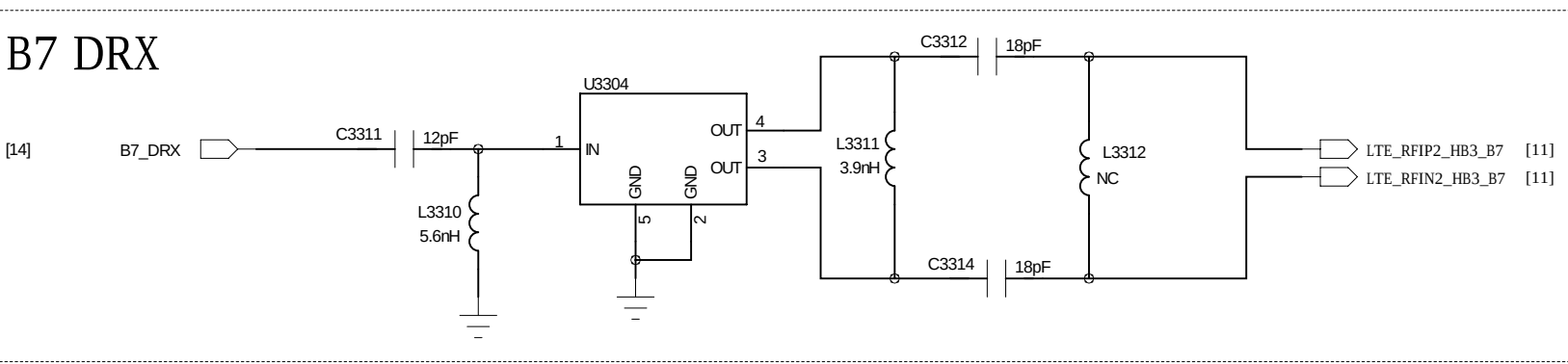
DRX ANT: 832~2690MHz



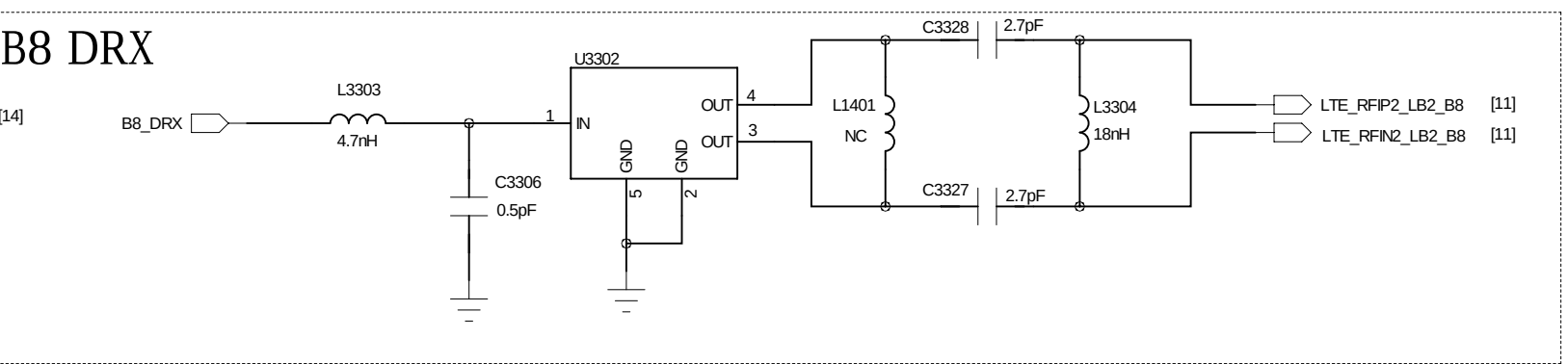
B3 DRX



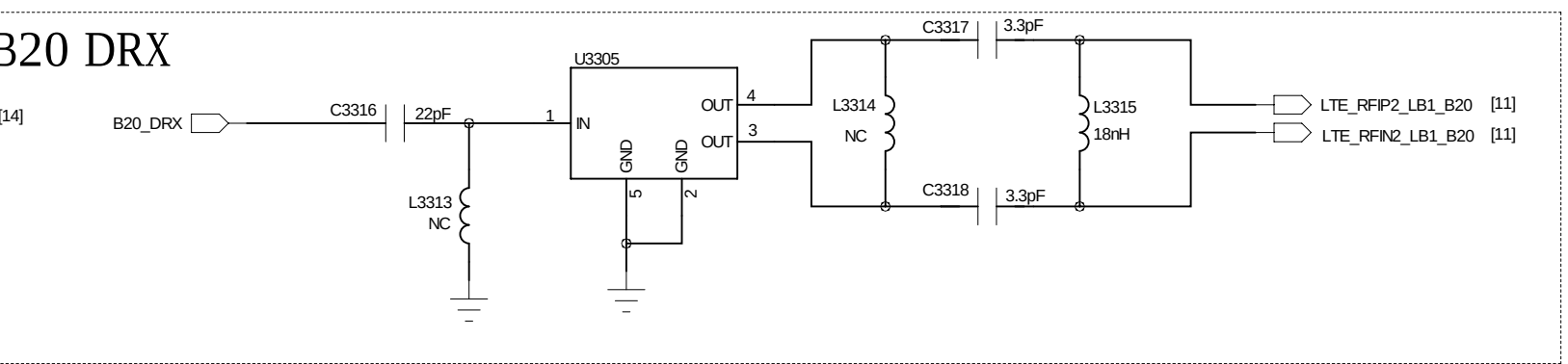
B7 DRX



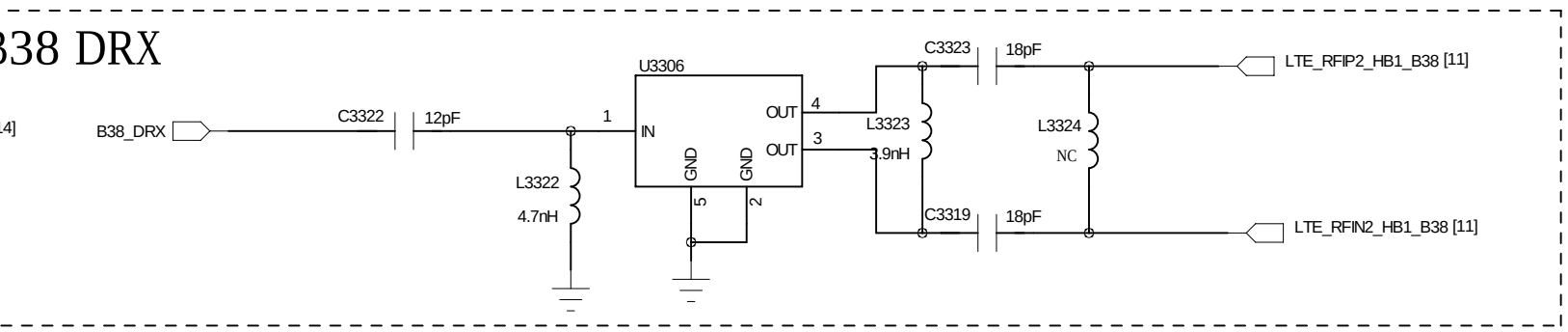
B8 DRX



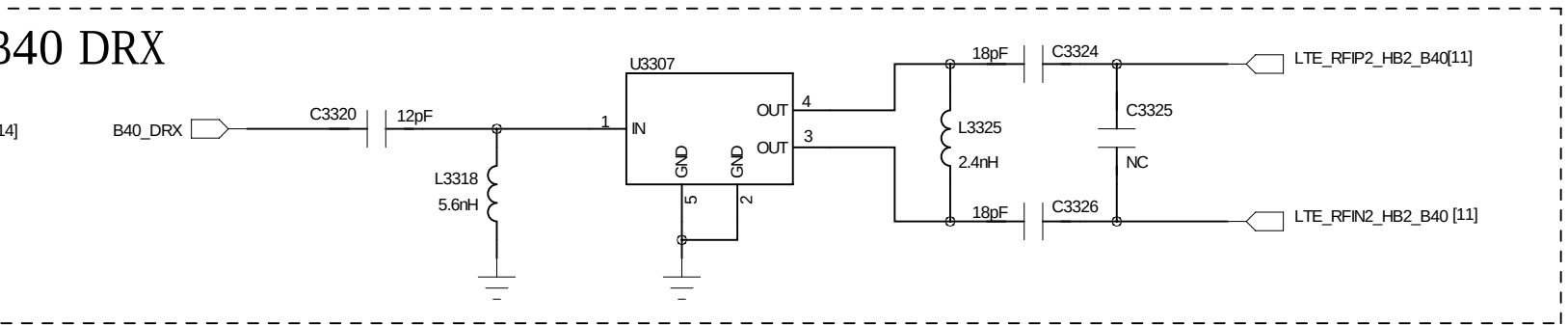
B20 DRX



B38 DRX



B40 DRX



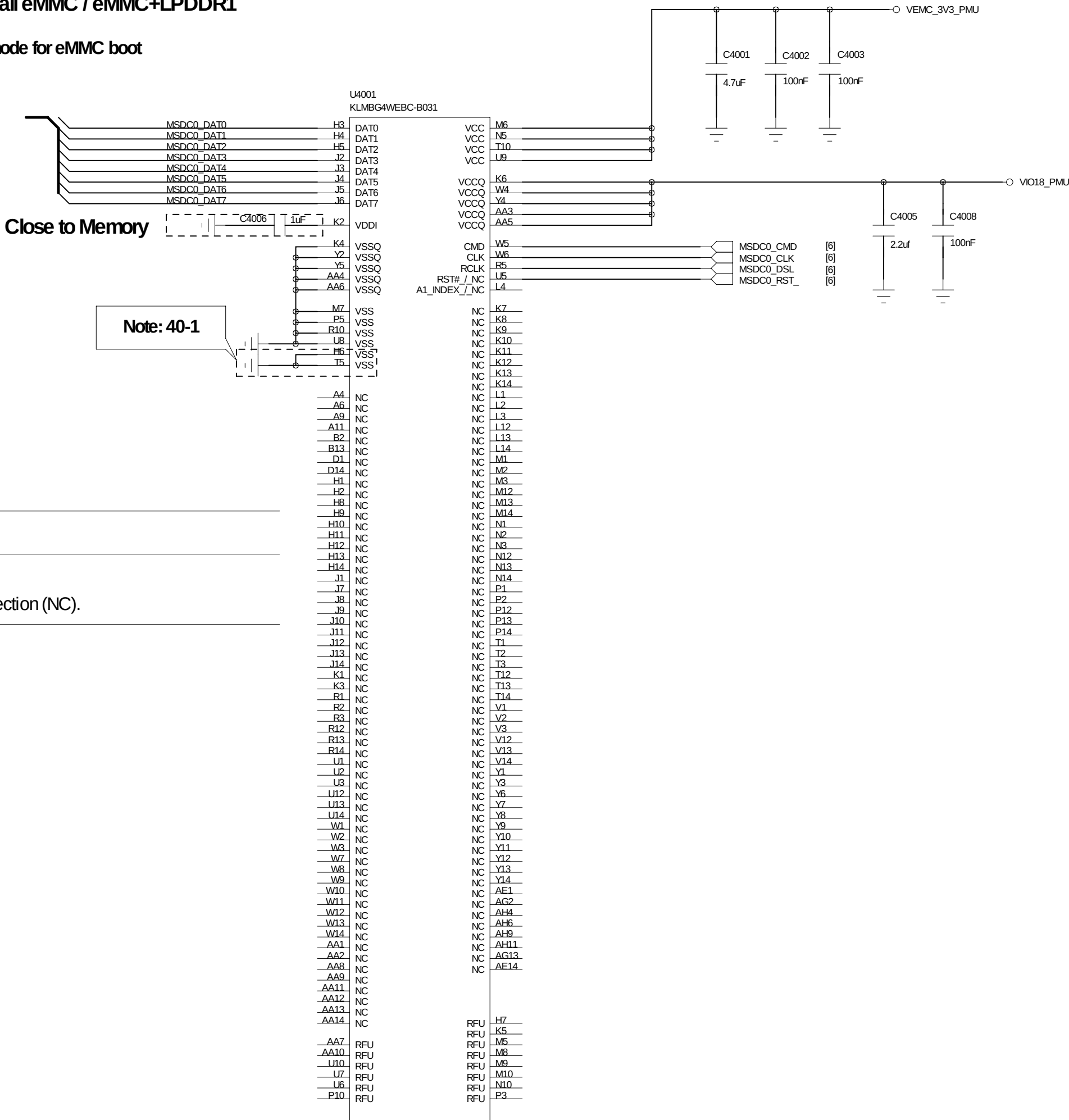
COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 33_RF_MT6169_RF_DRX		VERSION: V1.1	SHEET: 14 OF 22
CHECKED	< >	DATED	< >				

MEMORY_EMMC

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

169 ball eMMC / eMMC+LPDDR1

1 bit mode for eMMC boot



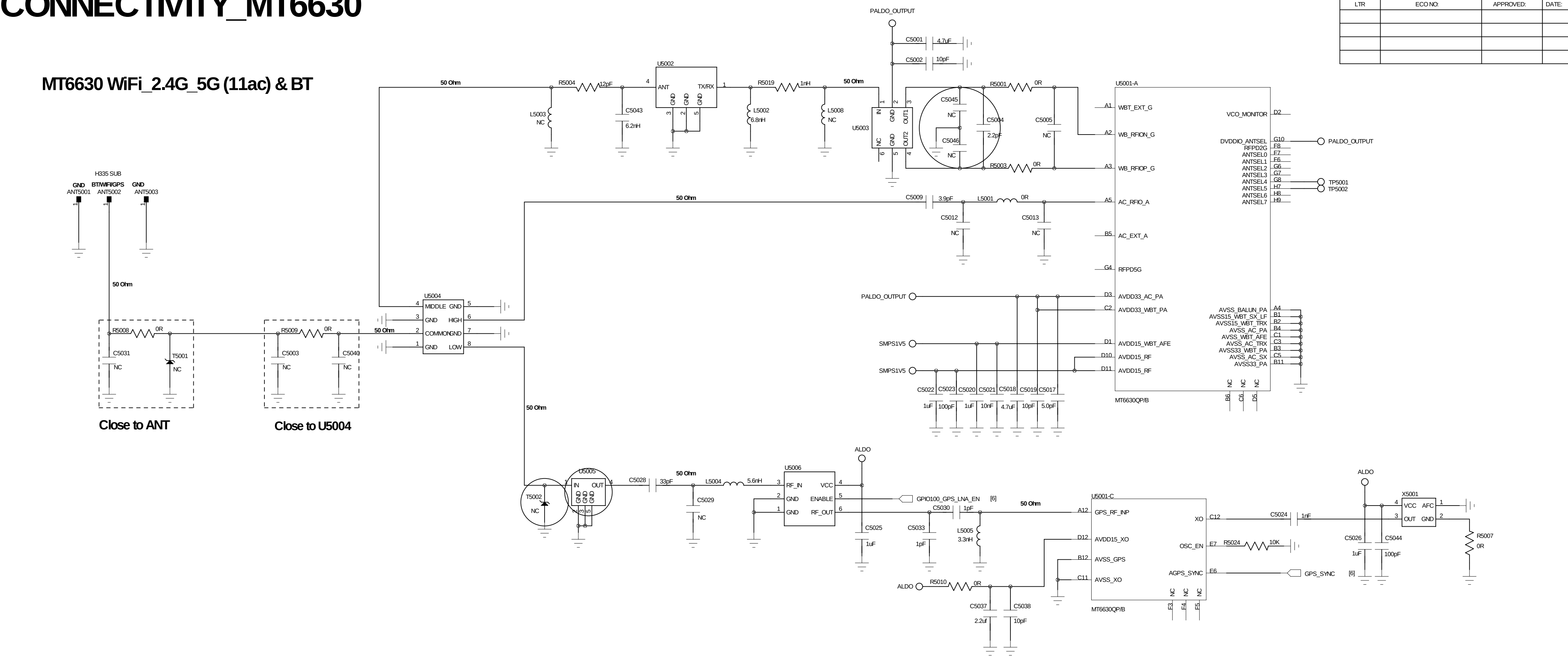
Schematic design notice of "40_MEMORY_eMMC" page.	
Note 40-1:	For eMMC 5.0, connect eMMC's H6 & T5 pin to GND. For eMMC 4.5, check eMMC's H6 & T5 is real no connection (NC).

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL:	H952	Modified Date:	2015-9-1
DRAWN	ZY/TS	DATED	20140512	TITLE:	40_MEMORY_EMMC	VERSION:	V1.1
CHECKED	< >	DATED	< >			SHEET:	15 OF 22

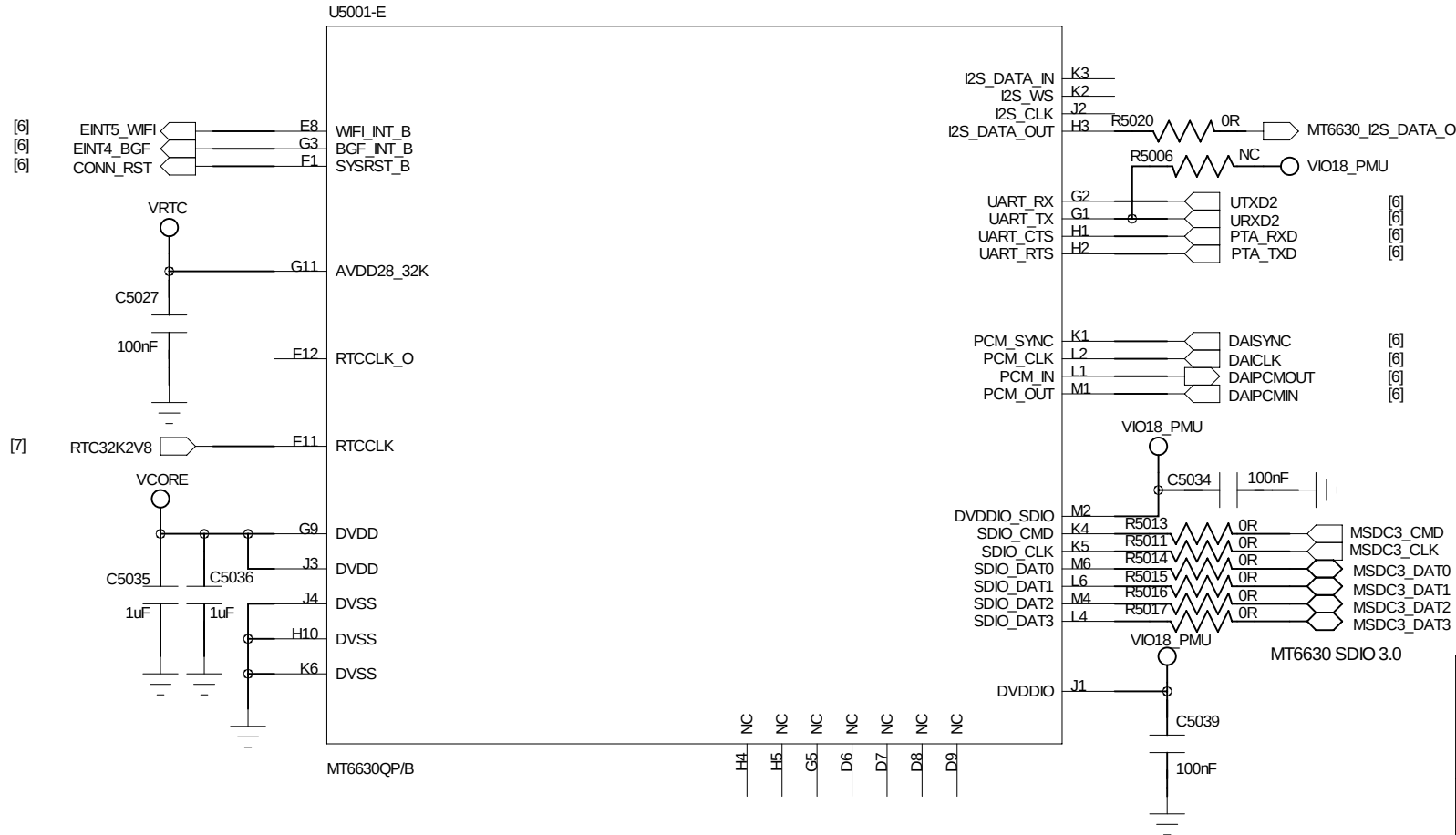
CONNECTIVITY_MT6630

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

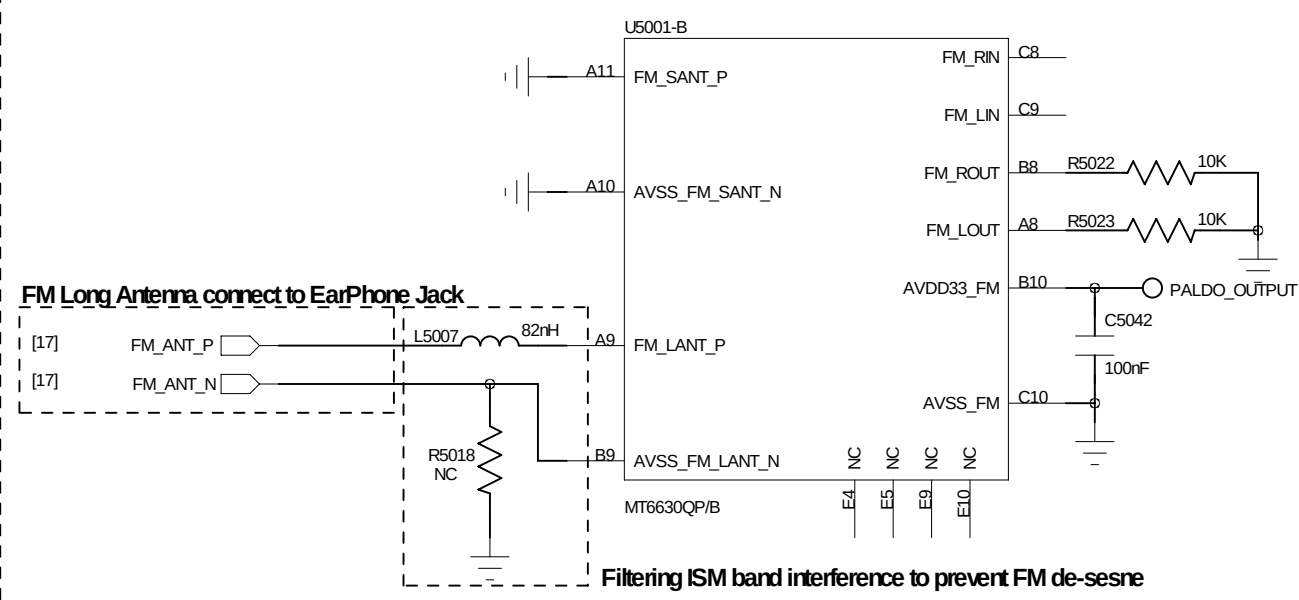
MT6630 WiFi_2.4G_5G (11ac) & BT



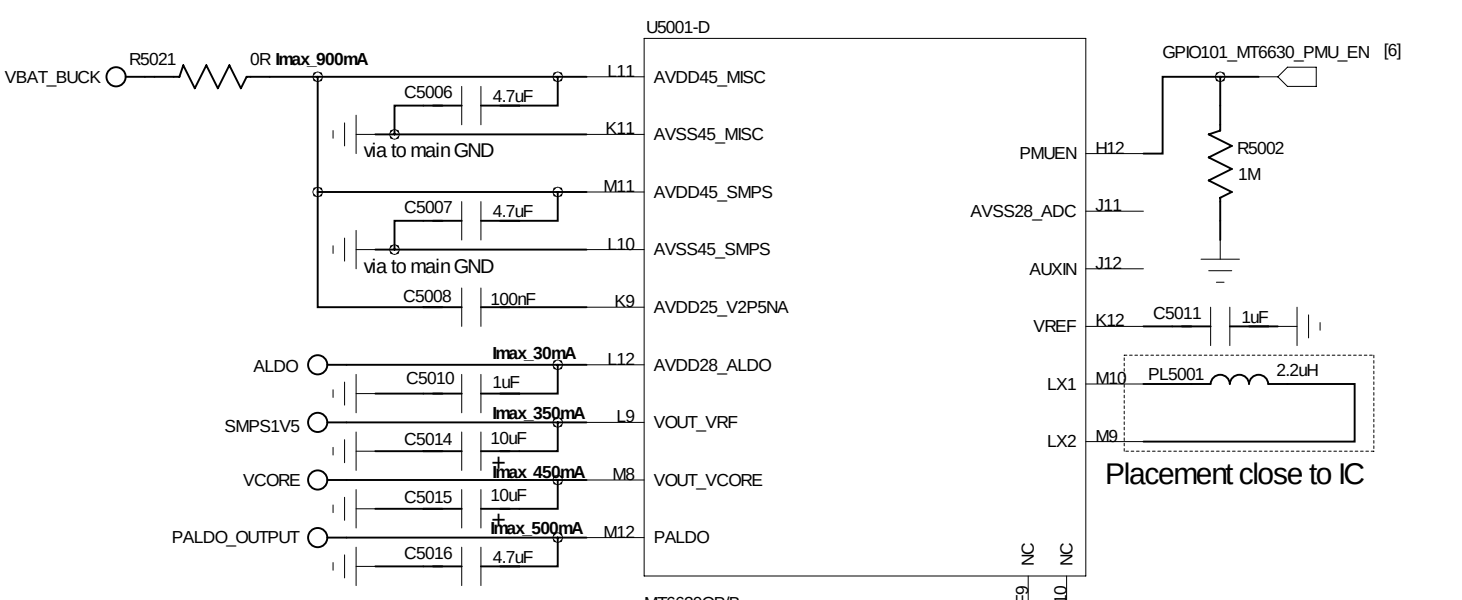
MT6630 Host Interface



MT6630_FM



MT6630 PMU

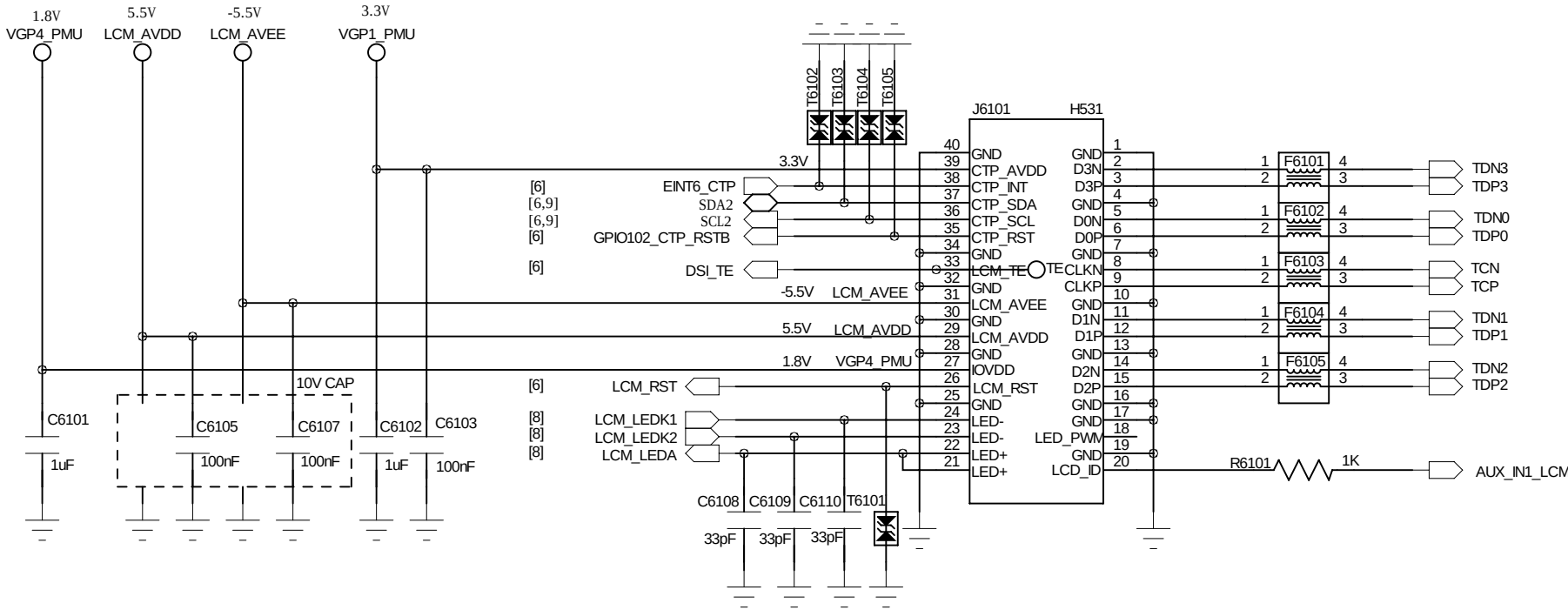


COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 50_4IN1_MT6630		VERSION: V1.1	SHEET: 16 OF 22
CHECKED	< >	DATED	< >				

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

5.46FHD
LCD_CTP_INTERFACE

CTP (GT910) I2C address: (Write:0x28, Read:0x29 or Write:0xBA, Read:0xBB)



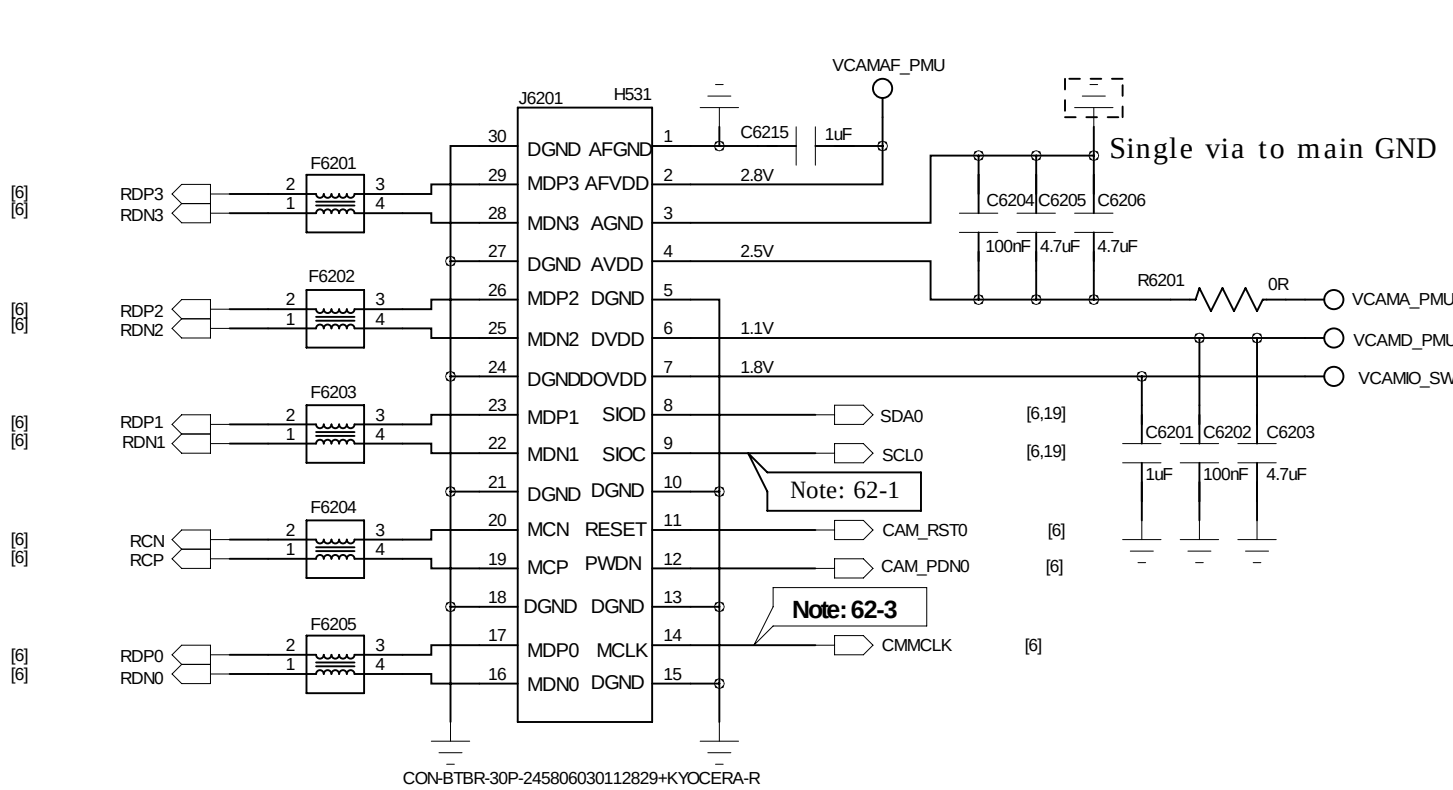
COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 61_PERI_LCD_CTP		VERSION: V1.1	SHEET: 18OF 22
CHECKED	< >	DATED	< >				

PERI_CAMERA

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

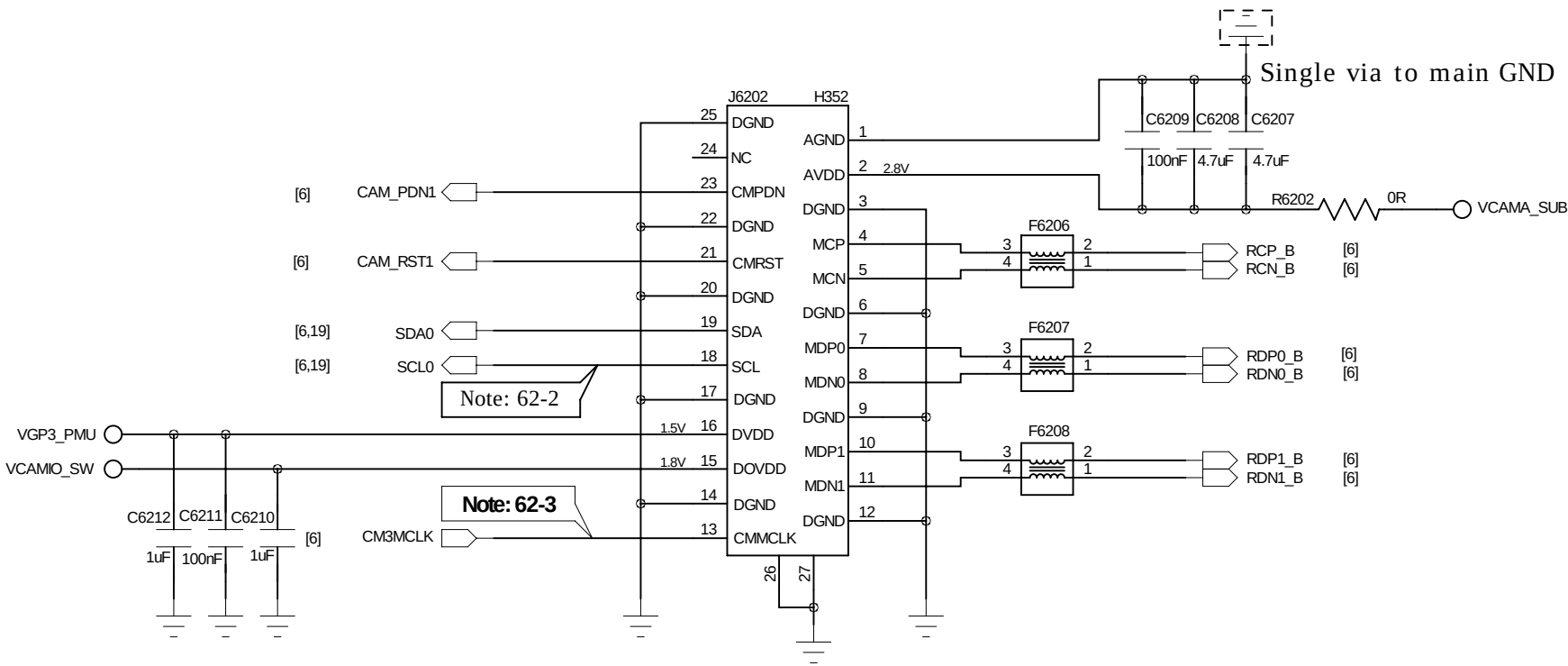
Rear Camera (20.7M+AF)

Rear camera (IMX230) I2C address: (Write:0x20, Read:0x21) or (Write:0x34, Read:0x35)
AF driver (BU64297GWZ) I2C address: (Write:0x18, Read:0x19)
EEPROM (GT24C16) I2C address: (Write:0xA0, Read:0xA1)

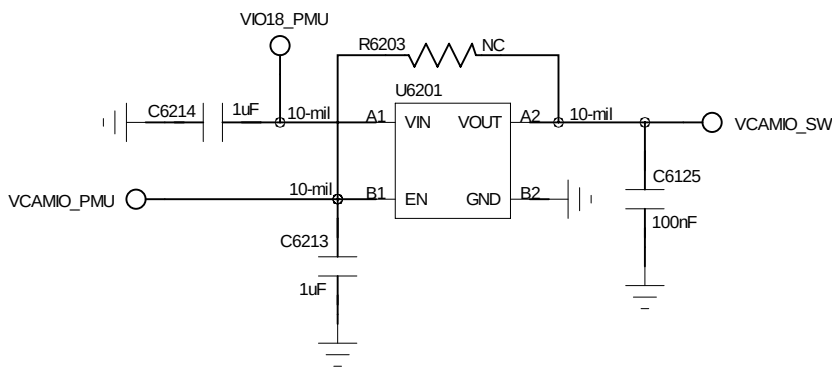


Front Camera (5M+FF)

Front camera (OV5648) I2C address: (Write:0x6C, Read:0x6D)



VCAM_IO 1.8V SW



Schematic design notice of "62_PERI_CAMERA_KEYPAD" page.

Note 62-1: The VCC of I2C_0 is pulled to "VCAM_IO_PMU".

Note 62-2: I2C control interface of front camera (with AF) must be assigned to I2C-3 bus when PIP/VIV feature be supported.

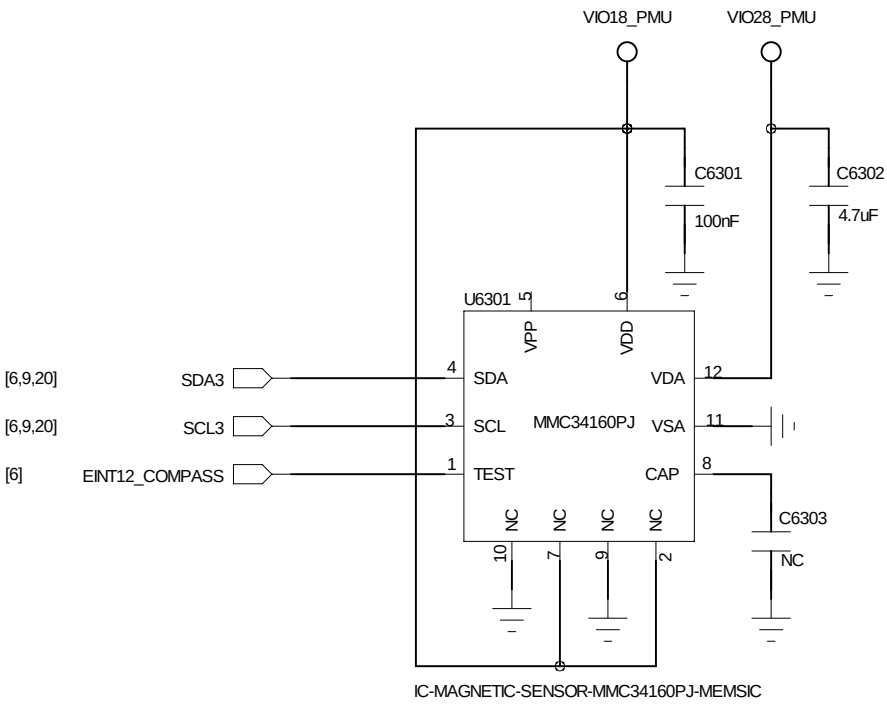
Note 62-3: Reserve a capacitor (27pF) on camera's MCLK and shunt it to GND to prevent GPS de-sense.

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952	Modified Date: 2015-9-1
DRAWN ZY/TS	DATED 20140512	TITLE: 62_PERI_CAMERA			VERSION: V1.1
CHECKED < >	DATED < >				SHEET: 19OF 22

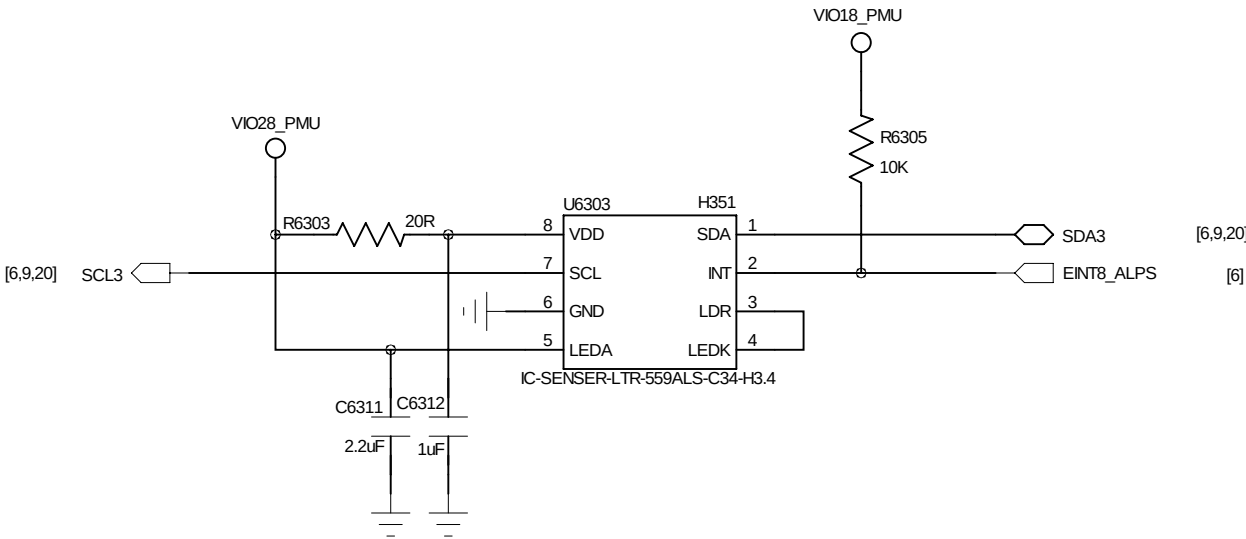
PERI_SENSORS

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

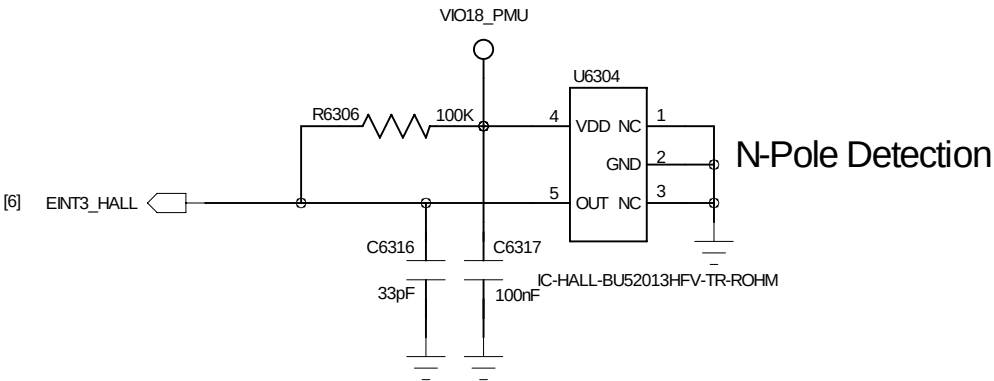
M-Sensor (COMPASS)
MMC35240PJ:I2C ADDRESS:0x60(Write)/0x61(Read)



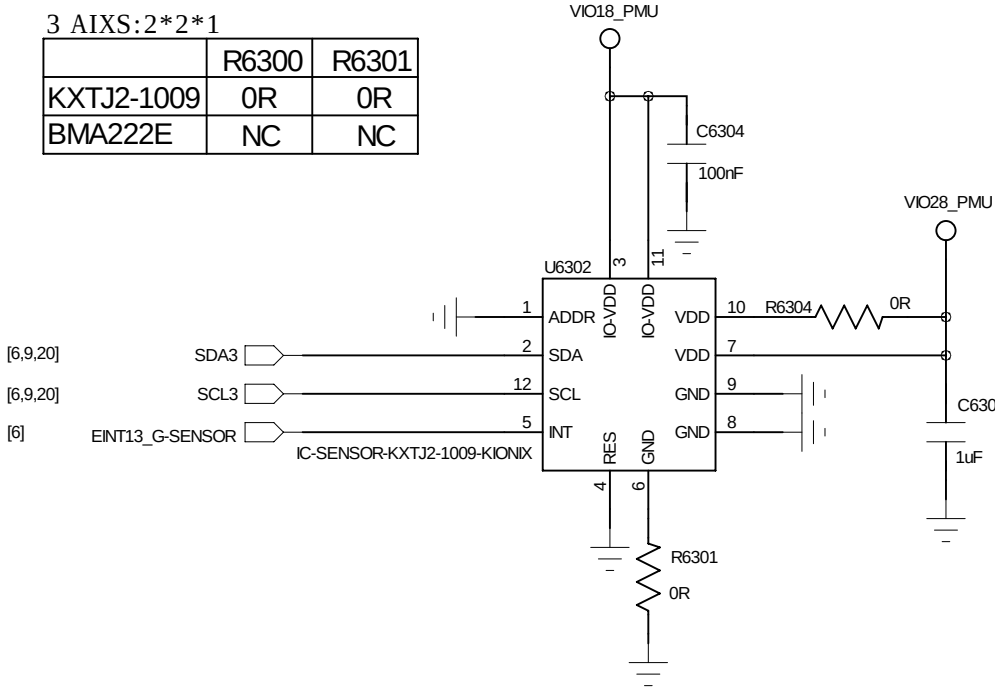
AL& PS Sensor
LTR-559ALS-034:I2C address: Write:0x46, Read:0x47



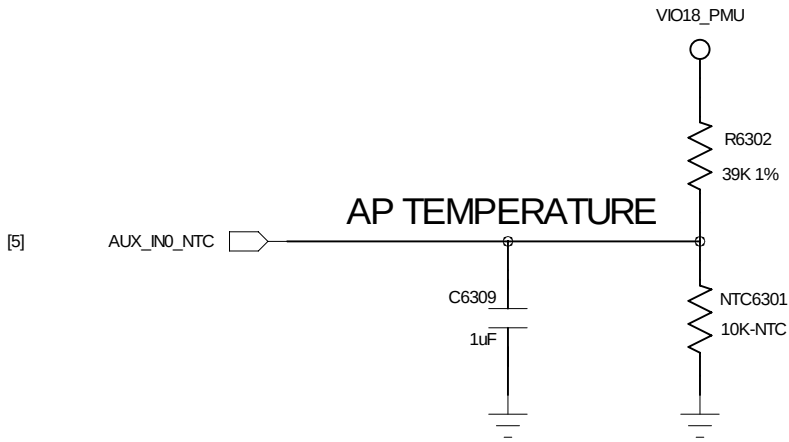
Unipolar HALL



G-Sensor (x)
Default KXTJ2-1009:I2C ADDRESS:0x1C(Write)/0x1D(Read)
BMA222E:I2C ADDRESS:0x30(Write)/0x31(Read)



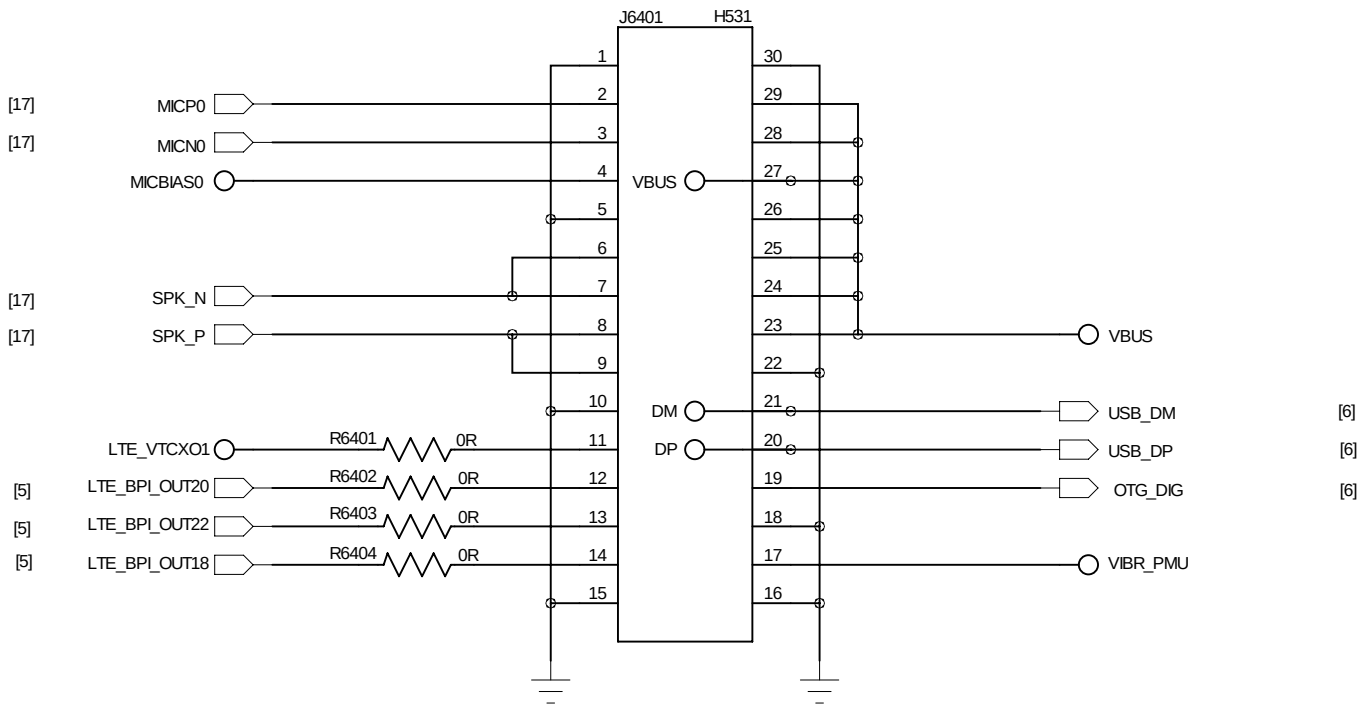
Thermister / To sense board level temperature



COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 63_PERI_SENSORS		VERSION: V1.1	SHEET: 20OF 22
CHECKED	< >	DATED	< >				

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

EXCHANGE IO

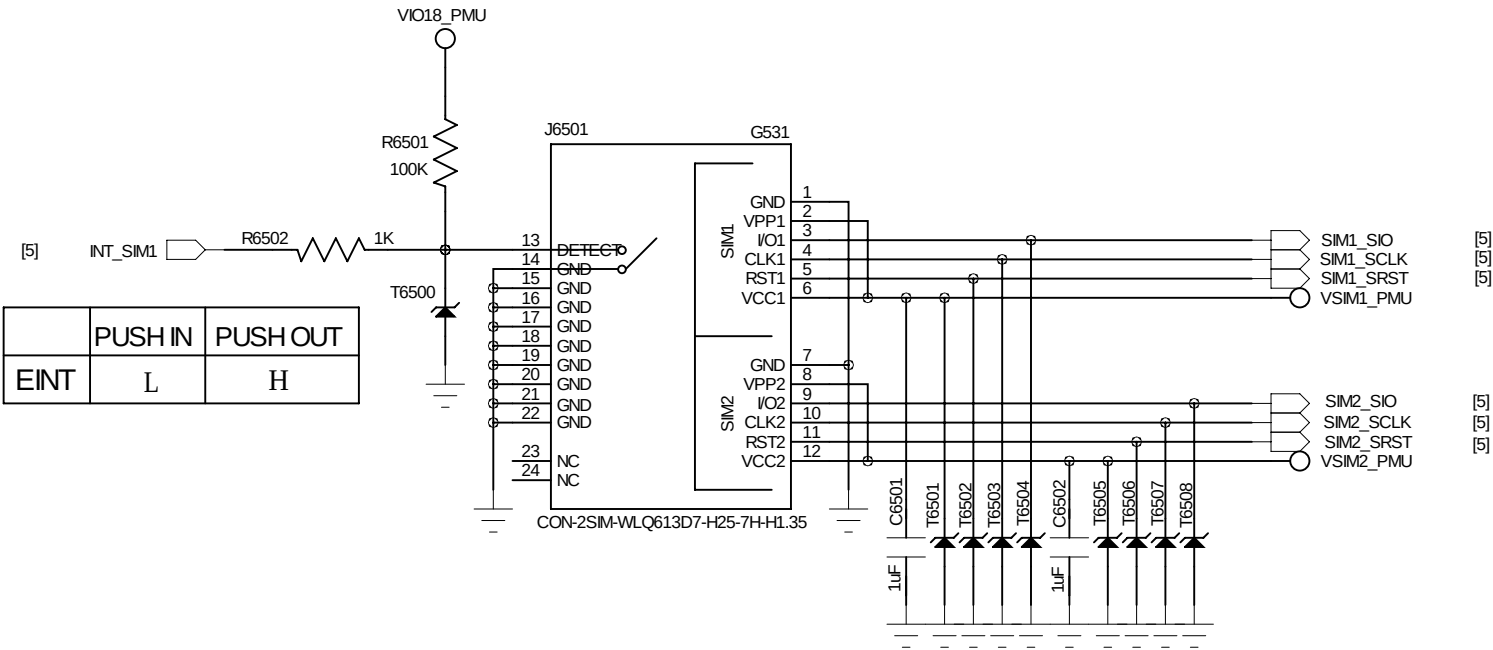


COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 64_PERI_EXCH_IO		VERSION: V1.1	SHEET: 21 OF 22
CHECKED	< >	DATED	< >				

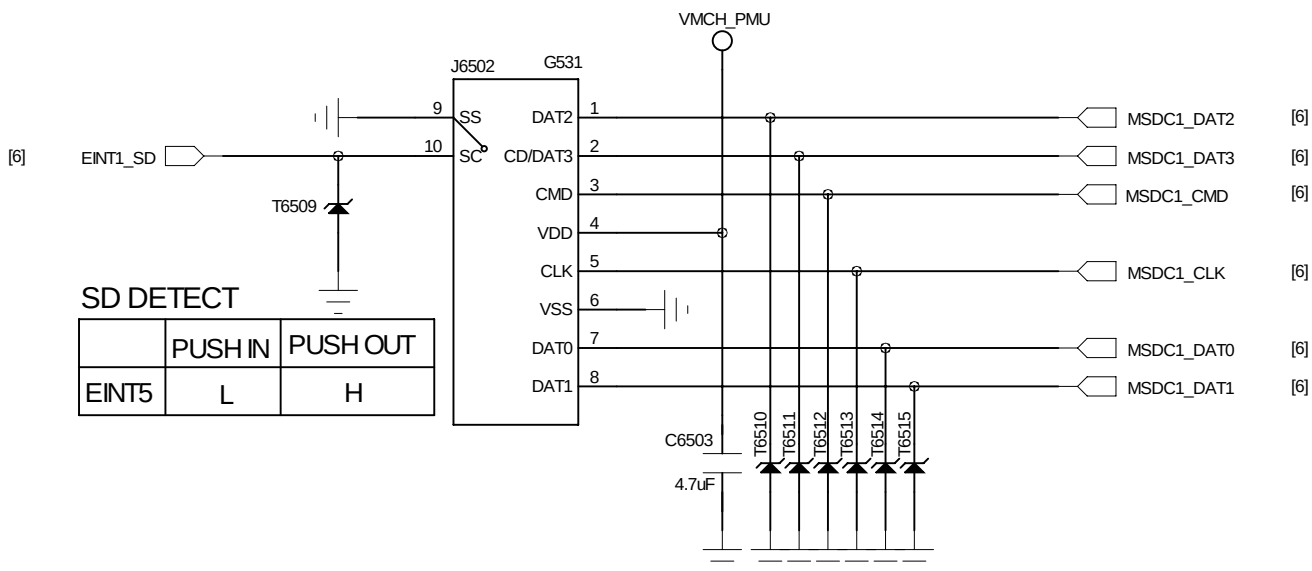
PERI_DUAL_SIM_KEYPAD

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

SIM CARD

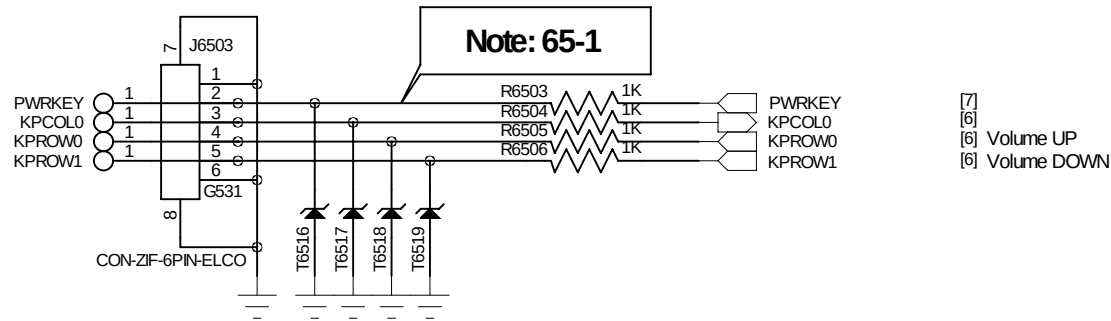


TF-CARD

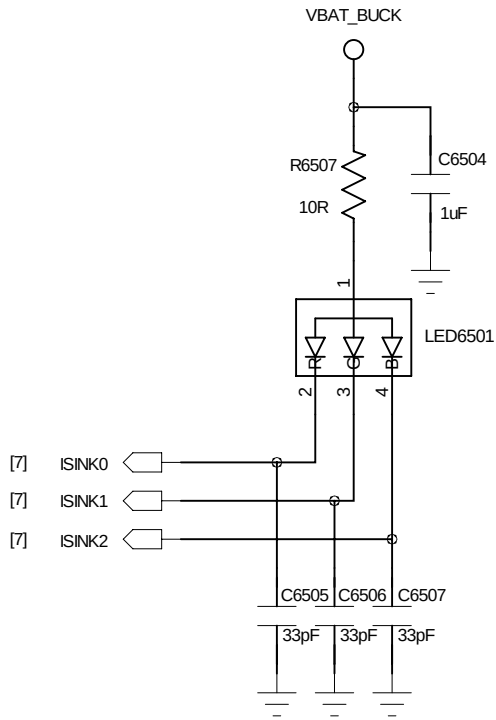


Power Key&Volume Key

DO NOT put pull-up resistor on PWRKEY



LED Indicators



Schematic design notice of "65_PERI_Dual_SIM_ICUSB_KEYPAD" page.

Note 65-1: DO NOT put pull-up resistor on PWRKEY

Note 65-3: Depends on system design to add ESD protection component or not.

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H952		Modified Date: 2015-9-1	
DRAWN	ZY/TS	DATED	20140512	TITLE: 65_PERI_DUAL_SIM_SD_KEYPAD		VERSION: V1.1	SHEET: 22 OF 22
CHECKED	< >	DATED	< >				